In the next section we relate the concept of linear independence to that of dimensionality.

3.3 BASES AND DIMENSIONALITY

Definition 3.5. A *basis* in a vector space V is a set $\{x_i\}$ of linearly independent vectors such that every vector in V is a linear combination of elements of $\{x_i\}$. A vector space is finite-dimensional if it has a finite set of basis vectors; otherwise it is infinite-dimensional.

The reader may want to prove that an equivalent alternative definition is that a basis is a maximal linearly independent subset of V.

If every vector in V is a linear combination of elements of $\{x_i\}$, we say that the set $\{x_i\}$ *spans* V, so a basis is a linearly independent set that spans V.

Examples of Bases.

1. In real three-dimensional space, the three vectors

$$\mathbf{e}_1 = (1, 0, 0)\ , \qquad \mathbf{e}_2 = (0, 1, 0)\ , \qquad \mathbf{e}_3 = (0, 0, 1)$$

form a basis. It is easy to show they are linearly independent, and that every vector, x, can be expanded as a linear combination of these three basis vectors, as

$$x = (\xi_1, \xi_2, \xi_3) = \sum_{i=1}^{3} \xi_i \mathbf{e}_i\ .$$

2. For the vector space of polynomials, P, the set $\{x_n\}$, where $x_n(t) = t^n$, $n = 0, 1, 2, \cdots$, is a basis, but not a finite one, so P is infinite-dimensional.

Theorem 3.2. Every vector in V has a unique representation as a linear combination of vectors of a fixed basis of V.

Proof. Let $\{x_i\}$ be a fixed basis in V. Then for every $x \in V$, x can be written as $x = \sum_i \alpha_i x_i$. The theorem says that the α_i are uniquely determined. To prove this, we assume the opposite, namely that there exists another representation of x, $x = \sum_i \beta_i x_i$. If this is the case, then by subtracting, we find that

$$0 = x - x = \sum_i (\alpha_i - \beta_i) x_i\ .$$

But since the x_i form a basis, they are linearly independent, and by the definition of linear independence, $(\alpha_i - \beta_i) = 0$ for all i. Thus $\alpha_i = \beta_i$ and the representation is unique. QED

Suppose that we are given a collection of linearly independent vectors $\{y_i\}$ where $y_i \in V$ for all i. If we consider the set of all linear combinations of the y_i, we see that this set of vectors is itself a vector space, every element of which lies in V. We say that this collection is the *subspace* of V *spanned* by the set $\{y_i\}$. In particular, V is a subspace of itself. Now it may happen that the set

$\{y_i\}$ is not a basis for V. We would expect that it is possible to add some vectors to the set $\{y_i\}$ and turn it into a basis for V. The following theorem shows that this is indeed the case.

Theorem 3.3. If V is a finite-dimensional vector space and $\{y_1, y_2, \cdots, y_m\}$ is a set of linearly independent vectors, then unless the y's are a basis, we may extend $\{y_1, y_2, \cdots, y_m\}$ by adding $y_{m+1}, \cdots, y_n$ so that the set $\{y_1, y_2, \cdots, y_n\}$ is a basis for V.

Proof. Let $\{x_1, x_2, \cdots, x_l\}$ be a basis for V. Then the set of vectors

$$S_1 = \{y_1, y_2, \cdots, y_m, x_1, x_2, \cdots, x_l\}$$

is linearly dependent. Hence, by Theorem 3.1, some member of S_1 is a linear combination of the vectors which precede it. It must be one of the x's since the y's are linearly independent. Call this element x_i and remove it from S_1 to form S_2:

$$S_2 = \{y_1, y_2, \cdots, y_m, x_1, x_2, \cdots, x_{i-1}, x_{i+1}, \cdots, x_l\}.$$

We claim that every vector in V can be expressed as a linear combination of elements of S_2. To see this, we note that the set S_1 spanned V, and the only element of S_1 missing from S_2 is x_i, which is a linear combination of elements of S_2; in fact, x_i is a linear combination of the vectors $\{y_1, y_2, \cdots, y_n, x_1, \cdots, x_{i-1}\}$.

Now, if the set S_2 is linearly independent, then we have found a basis for V; if not, there must be some x which is a linear combination of the elements of S_2 which precede it (again it must be an x, since the y's are linearly independent). We eliminate this x and form S_3. For the same reasons as before, every vector in V can be written as a linear combination of the elements of S_3. If S_3 is linearly independent, then it is a basis for V; if not, we continue this process until we find some S_n which is linearly independent. Since the S's have only a finite number of elements, this search can take only a finite number of steps. In this way, we find a basis which contains the set $\{y_1, y_2, \cdots, y_m\}$, as required. QED

Now the subspace spanned by the set $\{y_i\}$ is contained in V and the basis for this subspace, is contained in the basis for V. We would like to say that the subspace is of smaller dimension than V and to define the dimensionality of a space to be the number of elements in its basis. However, we do not yet know that this number is unique. The reader may think it obvious that all is well, but the demonstration of the uniqueness of the number of elements in a basis actually involves a rather subtle line of reasoning. Since the concept of dimensionality is an absolutely fundamental one, we shall go through this proof in detail.

Theorem 3.4. The number of elements in any basis of a finite-dimensional vector space is the same as in any other basis.

Proof. Let $\{x_1, x_2, \cdots, x_n\}$ and $\{y_1, y_2, \cdots, y_m\}$ be bases for V. Since n must either be equal to m or be different from m, we may assume without loss of

generality that $n \leq m$. The idea of the proof will be to show that $n \geq m$, which combined with the assumption $n \leq m$ leads to the conclusion $n = m$, which is just what we want to prove.

Now a set of basis vectors has two essential properties: (1) the vectors must be linearly independent; and (2) any vector in V must be a linear combination of the elements of the basis. We shall consider the set $\{y_i\}$ to have property (1) and the set $\{x_i\}$ to have property (2). Of course, each set has both the properties in question, but for the purposes of this proof we shall focus only on one property per set.

Consider the set

$$S_1 = \{y_m, x_1, x_2, \cdots, x_n\} .$$

Since $\{x_i\}$ has property (2), every vector in V is a linear combination of elements of S_1, and S_1 is a linearly dependent set. Hence by the argument of Theorem 3.3, there is some x, say x_i, which is a linear combination of the elements of S_1 which precede it. Let us remove this element and form T_1:

$$T_1 = \{y_m, x_1, \cdots, x_{i-1}, x_{i+1}, \cdots, x_n\}$$

with every vector in V a linear combination of elements of T_1, for the reasons given in Theorem 3.3. Now form

$$S_2 = \{y_{m-1}, y_m, x_1, \cdots, x_{i-1}, x_{i+1}, \cdots, x_n\} .$$

Clearly, S_2 is linearly dependent, and every vector in V is a linear combination of elements of S_2. Thus some x must be a linear combination of the preceding elements of S_2 (it must be an x rather than a y because the y's are linearly independent). Call this element x_j, and subtract it from S_2 to form T_2:

$$T_2 = \{y_{m-1}, y_m, x_1, \cdots, x_{i-1}, x_{i+1}, \cdots, x_{j-1}, x_{j+1}, \cdots, x_n\} .$$

Now we continue this process, in each step eliminating an x and inserting a y. After n steps, all the x's are gone and we have

$$T_n = \{y_{m-n+1}, \cdots, y_m\} .$$

But all the T_n have property (2); that is, all the T_n span V. Therefore if $m > n$, y_1 (or any y) is a linear combination of the other y's. But this contradicts the fact that the basis $\{y_i\}$ is linearly independent. The only way to avoid this contradiction is to have $n \geq m$. But we have assumed $n \leq m$, so we have the desired result. QED

If we had assumed $n \geq m$, then we would have reversed the roles of $\{y_i\}$ and $\{x_i\}$ in our proof and concluded that $m \geq n$, thereby obtaining the same result. Now we are free to make the definition that we are after:

Definition 3.6. The *dimension* of a finite-dimensional vector space V is the unique number of elements in a basis of V.

Note that in an n-dimensional vector space, any set containing more than n vectors is linearly dependent.

3.4 ISOMORPHISMS

Let F^n be the set of n-tuples of elements in a field F. In the first example given of a vector space, it was stated that F^n may serve as a prototype of the n-dimensional vector space V over F. We want to prove that every finite-dimensional vector space over a field F is "essentially the same as" some F^n.

Definition 3.7. Two vector spaces U and V (over the same field) are *isomorphic* if there exists a one-to-one correspondence between the vectors $x^{(i)}$ in U and $y^{(i)}$ in V, say $y^{(i)} = f(x^{(i)})$, such that

$$f(\alpha_1 x^{(1)} + \alpha_2 x^{(2)}) = \alpha_1 f(x^{(1)}) + \alpha_2 f(x^{(2)}) ,$$

and also a one-to-one correspondence between the vectors in V and the vectors in U.

Thus an isomorphism is a one-to-one correspondence preserving all linear relations, or, what is the same, preserving "algebraic structure." We shall examine this definition to see what is behind it, and in so doing generalize it.

Consider two closed systems $S = \{E, *\}$ and $S' = \{E', \cdot\}$, each with one operation. Assume that there is a one-to-one mapping (or function) f which assigns to each element $a, b, c, \cdots, \in E$ an element $f(a), f(b), f(c), \cdots, \in E'$, depicted thus:

$$\begin{array}{ll} \underline{E \qquad\quad} & \underline{E'} \\ a & \xrightarrow{f} f(a) \\ b & \xrightarrow{f} f(b) . \end{array}$$

Assume that $a*b = c$; then $c \to f(c) = f(a*b)$. If the algebraic structure of S and S' is to be identical, then it is necessary that a multiplication carried out on the left in S be paralleled by a multiplication of the corresponding elements on the right in S'. Since $a*b = c$, this would require the following equality to hold:

$$f(a) \cdot f(b) = f(c) = f(a*b) .$$

If this equation is satisfied for every a, b, then the systems $S = \{E, *\}$ and $S' = \{E', \cdot\}$ are said to be *isomorphic*.

As an example of two systems that are isomorphic—although at first glance they look very different—consider the six permutations [Eqs. (3.2)] and the six 2×2 matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I' , \quad \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix} = F' , \quad \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix} = D' ,$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = A' , \quad \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix} = C' , \quad \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} = B' . \tag{3.5}$$

We have not as yet dealt extensively with the multiplication of matrices, although the reader will recall that it was briefly considered in Section 1.4,

Eq. (1.22). There we encountered the multiplication of two arrays of numbers, (a_{ij}) and (b_{ij}), to give a third, (c_{ij}), using the rule

$$c_{ij} = \sum_{k=1}^{2} a_{ik} b_{kj} , \tag{3.6}$$

where the first subscript labels the row, and the second the column, of an element.

It is easy to verify that the six matrices of Eq. (3.5) form a non-abelian group with respect to matrix multiplication (Eq. 3.6) which we shall later prove is associative, and that the group multiplication table (3.4), is identical with that of the permutation group of three objects, if we prime all the elements. Thus the multiplicative group structure is maintained if we associate each element of the permutation group with the corresponding primed element of the group of matrices. Then, for example, under this mapping f, we have

$$\begin{aligned} B &\xrightarrow{f} B' \\ D &\xrightarrow{f} D' \\ B*D = A &\xrightarrow{f} A' = B' \cdot D' \end{aligned}$$

so that

$$f(B*D) = B' \cdot D' = f(B) \cdot f(D) .$$

A relation of the same form holds for any of the other elements, so these two systems are isomorphic.

In general, if two systems have the same multiplication table, they are isomorphic, or *structurally identical*. We can find out anything we want to know about one system by studying the other and then translating the results back into the language of the first. Suppose, for example, that we want to study a group of rotations. We look for a set of matrices which is isomorphic to the group of rotations; we can then study the matrix group to learn about the rotations. Such a group of matrices is called a *representation* of the original group. Thus the set of matrices $\{I', A', B', C', D', F'\}$ above is a matrix representation of the permutation group (sometimes called a *symmetric group*) for a set of three objects. We shall study group representations in Chapter 10.

In the Definition (3.7) of isomorphism given originally, as between two vector spaces, the operation $(+)$ is the same in both systems, U and V, which are both vector spaces. The example above makes it clear that the operation could be different in the two systems. Also, the linear operation of the multiplication of a vector by a scalar has been included at the same time as the linear operation of the addition of two vectors. The one condition could be broken down into two as follows:

$$f(x^{(1)} + x^{(2)}) = f(x^{(1)}) + f(x^{(2)}) , \qquad f(\alpha x) = \alpha f(x) .$$

We now prove the fundamental result regarding the algebraic structure of a vector space.

Theorem 3.5. Every n-dimensional vector space $V_n(F)$ over F is isomorphic to the space F^n.

Proof. Let $\{x_1, \cdots, x_n\}$ be any basis in V. Each vector $x \in V$ can therefore be written as

$$x = \sum_{i=1}^{n} \alpha_i x_i .$$

We know that the scalar coordinates α_i $(i = 1, \cdots, n)$ are uniquely determined (Theorem 3.2.). Now we propose the one-to-one correspondence $x \rightleftarrows (\alpha_1, \cdots, \alpha_n)$ between V and F^n; that is,

$$f(x) = (\alpha_1, \cdots, \alpha_n) ,$$

as an isomorphism of V and F^n. If

$$y = \sum_{i=1}^{n} \beta_i x_i ,$$

then

$$cx + dy = c \sum_{i=1}^{n} \alpha_i x_i + d \sum_{i=1}^{n} \beta_i x_i = \sum_{i=1}^{n} (c\alpha_i + d\beta_i) x_i ,$$

so that

$$f(cx + dy) = (c\alpha_1 + d\beta_1, c\alpha_2 + d\beta_2, \cdots, c\alpha_n + d\beta_n) .$$

Now we must compute $cf(x) + df(y)$ and compare this with $f(cx + dy)$:

$$cf(x) = c(\alpha_1, \alpha_2, \cdots, \alpha_n) = (c\alpha_1, c\alpha_2, \cdots, c\alpha_n) ,$$
$$df(y) = d(\beta_1, \beta_2, \cdots, \beta_n) = (d\beta_1, d\beta_2, \cdots, d\beta_n) ,$$

so that

$$cf(x) + df(x) = (c\alpha_1 + d\beta_1, \cdots, c\alpha_n + d\beta_n) = f(cx + dy) .$$

Therefore the one-to-one correspondence f is an isomorphism between $V(F)$ and F^n. Thus $V(F)$ and F^n are isomorphic. QED

It follows from this theorem that any two vector spaces of the same dimension over the same field are isomorphic, because both are isomorphic to F^n, and therefore isomorphic to each other. This theorem also means that from here on, we could restrict our attention to n-tuples, the prototypes of vectors, without loss of generality. But this necessitates the choice of a representation (or basis) and working with coordinates, and besides being tedious, it obscures important properties of vectors and vector spaces that are independent of the choice of basis. For the most part, therefore, we shall treat vector spaces without reference to a specific basis.

3.5 LINEAR TRANSFORMATIONS

Definition 3.8. A *linear transformation* or *linear operator* A on a vector space V is a correspondence that assigns to every $x, y \in V$, a vector Ax, $Ay \in V$ in such a way that

$$A(ax + by) = aAx + bAy ,$$

where a and b are scalars. The product C of two linear transformations A and B is defined by the equation $Cx \equiv A(Bx)$, and written $C = AB$. The sum of two linear transformations is defined by the equation $(A + B)x \equiv Ax + Bx$.

Examples of linear transformations.

1. Let $V(F)$ be the vector space of n-tuples of real numbers, and let a_{ij} $(i, j = 1, 2, \cdots, n)$ be a set of n^2 real numbers. If we define $y = Ax$ by

$$y_i = \sum_{j=1}^{n} a_{ij}x_j ,$$

then A is a linear transformation. If for the case $n = 2$ we take $a_{11} = a_{22} = \cos\phi$, and $a_{12} = -a_{21} = \sin\phi$, then $A \equiv (a_{ij})$ becomes the rotation matrix of Eq. (1.14). This linear transformation corresponds to rotating the vector x through an angle ϕ to produce the new vector y.

2. Two special linear transformations O and I are defined by:

$$Ox \equiv 0 , \qquad Ix \equiv x .$$

3. The differentiation and multiplication linear transformations, D_x and X, are defined by:

$$D_x f(x) \equiv \frac{df(x)}{dx} \qquad \text{and} \qquad Xf(x) \equiv xf(x) .$$

Strictly speaking, D_x is an operator in the sense of Definition 3.8 only if, given a function in V, the derivative of this function is also in V. For example, if V is the space of continuous functions, then $|x| \in V$, but $d/dx\,|x|$ is not in V. Thus D_x is not defined everywhere in the space.

The product of D_x and X produces the following result when applied to a vector $f(x)$:

$$(D_xX)f(x) = D_x(Xf(x)) = \frac{d}{dx}(xf(x)) = f(x) + x\frac{df(x)}{dx} .$$

On the other hand, the product of X and D_x gives

$$(XD_x)\,f(x) = X(D_x f(x)) = x\frac{df(x)}{dx} .$$

Therefore

$$(D_xX)\,f(x) \neq (XD_x)\,f(x) ,$$

or for short, omitting the operand,

$$D_x X \neq X D_x .$$

In fact,

$$(D_x X - X D_x) f(x) = f(x) , \qquad \text{or} \qquad D_x X - X D_x = I .$$

This particular pair of noncommuting operators plays a central role in quantum mechanics, where the operator associated with the x-component of momentum is $p_x = -i\hbar D_x$, and X is the position operator. (There is a slight embarrassment with the notation here. Although we have written the multiplication operator as X above, it is customary among physicists to write this operator simply as x, since it has the effect of multiplying the operand by x. We shall henceforth abide by this convention.) Thus we write

$$[p_x x - x p_x] f(x) = -i\hbar f(x) , \qquad \text{or} \qquad p_x x - x p_x = -i\hbar .$$

The operator $(p_x x - x p_x)$ is the called the *commutator* of p_x and x. It is written as $[p_x, x]$. The rule $[p_x, x] = -i\hbar$ is taken as an axiom of quantum mechanics.

4. In the space P_m of polynomials of degree $\leq m$, the linear transformation D is given by

$$D p_m(x) = \frac{d}{dx} p_m(x) .$$

Let

$$D^1 = D , \qquad D^2 = DD , \qquad D^n = \underbrace{DD \cdots D}_{n \text{ times}} .$$

Then if $n > m$, $D^n p_m = 0$ for all polynomials in P_m, so that $D^n = 0$. Thus even if an operator is not the zero operator, some power of it may be the zero operator!

Most of the formal algebraic properties of numerical multiplication, with the notable exception of commutativity, are valid in the algebra of linear transformations. Thus we have:

a) $AO = OA = O$,
b) $AI = IA = A$,
c) $A(B + C) = AB + AC$,
d) $A(BC) = (AB)C$,
e) $(aA) = a(A)$, where a is a scalar.

We prove (c):

$$\begin{aligned}[A(B + C)]x &= A[(B + C)x] = A(Bx) + A(Cx) \\ &= (AB)x + (AC)x = (AB + AC)x ,\end{aligned}$$

where we have used the definitions of product and sum in Definition (3.8). Thus the *sum* of two linear transformations is commutative and associative because addition in V is so. Property (d) above says that the product is also

associative. The reader may now prove that the set of all linear transformations on a vector space $V(F)$ is itself a vector space over F.

3.6 THE INVERSE OF A LINEAR TRANSFORMATION

The following definition is designed to cover the infinite dimensional case as well as the finite dimensional case.

Definition 3.9. If a linear transformation A has both the following two properties, then there exists an *inverse* to A, denoted by A^{-1}, and we say that A is *invertible*.

a) $x \neq y \Longrightarrow Ax \neq Ay$ (or equivalently, $Ax = Ay \Longrightarrow x = y$).

b) For every $y \in V$ there exists an $x \in V$ such that $Ax = y$.

We construct A^{-1} as follows: If y_0 is any vector, we may [by (b)] find an x_0 such that $y_0 = Ax_0$. This x_0 is in fact uniquely determined, since if $Ax_0 = y_0$ and $Ax = y_0$, then $Ax_0 = Ax \Longrightarrow x_0 = x$ [by (a)]. We define $A^{-1}y_0 \equiv x_0$. Now, is A^{-1} a *linear* transformation? To establish linearity, we must prove

$$A^{-1}(\alpha x + \beta y) = \alpha A^{-1}x + \beta A^{-1}y .$$

If $Ax = u$ and $Ay = v$, then linearity of A implies

$$A(\alpha x + \beta y) = \alpha u + \beta v ,$$

so that

$$A^{-1}(\alpha u + \beta v) = \alpha x + \beta y = \alpha A^{-1}u + \beta A^{-1}v ,$$

and the inverse is linear. It follows immediately from the definition of the inverse that if A is invertible,

$$AA^{-1} = A^{-1}A = I .$$

The following theorem shows that whenever a linear transformation plays the role of A^{-1} in the above equation, then it must be the inverse of A.

Theorem 3.6. If A, B, and C are linear transformations such that $AB = CA = I$, then A is invertible and $A^{-1} = B = C$.

Proof. To prove A is invertible, we must prove the two conditions, (a) and (b), of Definition (3.9).

a) If $Ax = Ay$ then $CAx = CAy$, which implies $x = y$ (since $CA = I$).

b) If y is any vector and $x = By$, then $y = ABy = Ax$ (since $AB = I$).

Therefore A is invertible.

Now multiply $AB = I$ on the left and $CA = I$ on the right by A^{-1} to get $A^{-1} = B = C$. QED

Neither of the two conditions, $AB = I$ or $CA = I$, is alone sufficient to ensure the invertibility of A, unless we are in a finite-dimensional vector space,

or unless we also know that B or C is unique (see Problem 10); in these cases, one such condition suffices on any vector space.

To illustrate these statements, consider the two operators D_x and S operating on vectors of the infinite-dimensional vector space P, where

$$D_x p(x) \equiv \frac{d}{dx} p(x) , \qquad Sp(x) \equiv \int_0^x p(t)\, dt .$$

Although $D_x S = I$, neither D_x nor S is invertible. D_x violates condition (a), since, for instance, $D_x(x^2 + 3) = D_x(x^2 + 1)$; S violates condition (b), for if $p_0 = x^2 + 1$, there exists no $p(t)$ such that $\int_0^x p(t)\, dt = p_0$.

Theorem 3.7. Let A be a linear transformation defined on a finite-dimensional vector space V. If $Ax = 0$ implies that $x = 0$ then A is invertible.

Proof. If $Ax = Ay$, then $Ax - Ay = A(x - y) = 0$, which, by hypothesis, implies $(x - y) = 0$, or $x = y$, hence condition (a) for invertibility is satisfied. To prove condition (b), let $\{x_1, \cdots, x_n\}$ be a basis in V. (Note that we can pick a *finite* basis.) If it can be shown that $\{Ax_1, \cdots, Ax_n\}$ is also a basis, then every vector $y \in V$ can be written in the form

$$y = \sum_{i=1}^{n} \alpha_i A x_i = A\Big(\sum_{i=1}^{n} \alpha_i x_i\Big) ,$$

fulfilling condition (b). Thus we ask: Is $\{Ax_1, \cdots, Ax_n\}$ also a basis? It is a set of n vectors in an n-dimensional vector space, so it will be a basis if the set is linearly independent. To prove linear independence, we assume that

$$\sum_i \alpha_i A x_i = 0 .$$

It follows that

$$A\Big(\sum_{i=1}^{n} \alpha_i x_i\Big) = 0 \Longrightarrow \sum_{i=1}^{n} \alpha_i x_i = 0 ,$$

by hypothesis. The linear independence of the x_i now implies $\alpha_i = 0$ for all i, so the set $\{Ax_i\}$ is linearly independent and hence does form a basis. QED

The converse also holds: If A is invertible, then (obviously) $Ax = 0$ implies $x = 0$.

It should be noted that the condition of Theorem 3.7, that $Ax = 0 \Longrightarrow x = 0$, is equivalent to condition (a) of Definition 3.9. Thus Theorem 3.7 shows that on a *finite* dimensional space, condition (a) alone suffices for invertibility.

With the help of Theorem 3.6 the reader should have no trouble in supplying the simple proofs of the statements comprising the following theorem.

Theorem 3.8. If A and B are invertible, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$. If A is invertible and $\alpha \neq 0$, then αA is invertible and $(\alpha A)^{-1} = (1/\alpha)A^{-1}$. If A is invertible, then A^{-1} is invertible and $(A^{-1})^{-1} = A$.

3.7 MATRICES

Just as abstract vectors can be represented in terms of coordinates relative to a particular basis, so abstract linear transformations can be represented as matrices in terms of a given basis. We now develop the connection between linear transformations and matrices.

Definition 3.10. Let V be an n-dimensional vector space, with $X = \{x_1, \cdots, x_n\}$ a basis in V, and let A be a linear transformation mapping V into itself.

Since every vector is a linear combination of the $\{x_i\}$, we have in particular

$$Ax_j = \sum_{i=1}^{n} a_{ij}x_i\,, \qquad \text{for } j = 1, \cdots, n\,. \tag{3.7}$$

The set of n^2 scalars (a_{ij}) is the *matrix* of A relative to the basis X (or in the "coordinate system" X).

$$(a_{ij}) \equiv \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdot & & & \cdot \\ \cdot & & & \cdot \\ \cdot & & & \cdot \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \equiv [A] \equiv A \qquad \text{(by sloppy convention)}\,.$$

That is, every transformed vector Ax_j (for all j) can be reexpressed in terms of the basis vectors x_i by n coefficients, as previously. But there are n basis vectors, hence n transformed vectors Ax_j $(j = 1, \cdots, n)$, and hence n^2 coefficients, a_{ij}.

Note carefully the order of indices in the equation that defines the a_{ij}. The rule is: write Ax_j as a linear combination of $(x_1, \cdots, x_n)$, and write the coefficients so obtained as the jth *column* of the matrix A. The first index on $a_{ij} \equiv [A]_{ij}$ is the row index, the second the column index. The location of the elements in the matrix representation of A will depend on the ordering of the vectors in the basis, but we are free to adopt a fixed ordering of basis vectors if it becomes desirable to associate a uniquely-arranged array of elements with a linear transformation. We say that two matrices (a_{ij}) and (b_{ij}) are equal if and only if $a_{ij} = b_{ij}$, for all i and j.

In Section 1.4, Eq. (1.13), we expanded one set of basis vectors, $\mathbf{e}'_j$, in terms of another set, $\mathbf{e}_i$, to define a matrix. There we had the equation

$$\mathbf{e}'_j = \sum_i a_{ji}\mathbf{e}_i\,.$$

The $\mathbf{e}_i$ played the role that the x_i play now. The x_i in Chapter 1 referred to components of the vectors $\mathbf{x}$, whereas now the x_i are themselves vectors in an abstract vector space. However, the convention of Chapter 1 that results in the above equation is not our present one. In terms of the convention of this chapter, we should have written in Chapter 1,

$$\mathbf{e}'_j = \sum_i a_{ij}\mathbf{e}_i\,.$$

This notational inconsistency, and the reasons for retaining it, were discussed in Chapter 1. All the equations of Chapter 1 may be converted to our present convention by simply replacing a_{ij} by a_{ji}.

Now let us see how the application of a linear transformation A to a vector x looks in terms of the representation of these abstract objects with respect to a basis, $X = \{x_1, \cdots, x_n\}$. For any $x \in V$,

$$x = \sum_{j=1}^{n} \alpha_j x_j .$$

And the matrix of the linear transformation is defined by

$$Ax_j = \sum_{i=1}^{n} a_{ij} x_i .$$

Therefore

$$y = Ax = A \sum_{j=1}^{n} \alpha_j x_j = \sum_{j=1}^{n} \alpha_j A x_j = \sum_{j=1}^{n} \alpha_j \sum_{i=1}^{n} a_{ij} x_i = \sum_{i,j=1}^{n} a_{ij} \alpha_j x_i . \quad (3.8)$$

But the vector y, produced by A operating on x, may also be represented in terms of the basis X:

$$y = \sum_{i=1}^{n} \beta_i x_i .$$

Subtracting this from Eq. (3.8), we have

$$\sum_{i=1}^{n} \left[\beta_i - \sum_{j=1}^{n} a_{ij} \alpha_j \right] x_i = 0 ,$$

which, from the linear independence of the basis, implies

$$\beta_i = \sum_{j=1}^{n} a_{ij} \alpha_j .$$

This conforms exactly with our earlier specialized result, Eq. (1.16c), the β_i corresponding to the x'_i. This equation is often written symbolically as

$$y = Ax , \quad (3.9)$$

where y and x stand for the coordinate representatives of the vectors y and x, and A stands for the matrix representing the linear transformation A, all referred to the same basis X. A more accurate notation than that in Eq. (3.9) is

$$[\beta] = [A][\alpha] , \quad (3.10)$$

where $[\beta]$ and $[\alpha]$ are matrices of n rows and one column, called *column matrices*, or *column vectors*, defined by

$$[\beta]_{i1} = \beta_i , \qquad i = 1, \cdots, n ,$$

and similarly for α, and $[A]$ is the matrix of A. In practice, however, one usually falls back on the sloppier, simple notation of Eq. (3.9).

Row vectors are matrices of one row and n columns. Some authors distinguish carefully between row and column vectors but we shall not; the context always enables one to do so if necessary.

Thus far, we have discussed only square matrices, i.e., those having the same number of rows and columns. The reason for this is that the operators, A, in question are defined to take vectors in V into vectors in V, that is, to map V into itself. However, we could equally well imagine a linear operator B which maps V into some other vector space U. If V and U happen to be of different dimensionalities, the matrix which represents B will not be square.

Let $\{x_i\}$ be a basis for V and $\{y_i\}$ be a basis for U; then Bx_j is a vector in U so we may write

$$Bx_j = \sum_{i=1}^{m} b_{ij} y_i \qquad j = 1, 2, \cdots, n ,$$

where we have taken V to be n-dimensional and U to be m-dimensional, so that (b_{ij}) is an $m \times n$ matrix, having m rows and n columns. We leave it to the reader to show that if x is a vector in V and y is a vector in U such that $y = Bx$, and if

$$y = \sum_{i=1}^{m} \beta_i y_i , \qquad x = \sum_{i=1}^{n} \alpha_i x_i ,$$

then

$$\beta_i = \sum_{j=1}^{n} b_{ij}\alpha_j , \qquad i = 1, 2, \cdots, m ,$$

which we again write as

$$[\beta] = [B][\alpha]$$

in analogy with Eq. (3.10).

Example. We shall examine the form of the matrices of the differentiation and multiplication operators, D and X. Earlier we remarked that in talking about linear operators, one had to be sure that they were defined on the space in question. D, for example, is certainly defined on the space P_n of polynomials of degree $\leq n$. However, it turns every element of P_n into an element of P_{n-1}; this is why D is not invertible when considered as an operator mapping P_n into P_n.

Since P_{n-1} is a subspace of P_n, we can consider D as either an operator which maps P_n into P_{n-1}, or P_n into P_n. In the former case, D will be represented by an $n \times (n + 1)$ matrix, in the latter by an $(n + 1) \times (n + 1)$ matrix, [By an unfortunate convention, P_n is an $(n + 1)$-dimensional space.]

We use the obvious bases

$$y_j(t) = t^{j-1} , \qquad j = 1, 2, \cdots, n \qquad \text{for } P_{n-1};$$
$$x_j(t) = t^{j-1} , \qquad j = 1, 2, \cdots, n + 1 \qquad \text{for } P_n .$$

Then

$$Dx_j = (j-1)t^{j-2} = (j-1)y_{j-1} = \sum_{i=1}^{n} d_{ij}y_i\,, \qquad j = 1, 2, \cdots, n+1\,.$$

Therefore $d_{ij} = j - 1$ if $i = j - 1$, and $d_{ij} = 0$ otherwise. Thus

$$[D] = (d_{ij}) = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 2 & 0 & & 0 & 0 \\ 0 & 0 & 0 & 3 & & 0 & 0 \\ \vdots & & & & & & \vdots \\ 0 & 0 & 0 & 0 & & n-1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & n \end{bmatrix}. \tag{3.11}$$

Thus $[D]$ is an $n \times (n+1)$ matrix. If we wanted to think of D as mapping P_n into P_n, we would represent D by a matrix obtained from the one above by adding an $(n+1)$th row of zeroes.

The operator X can only be represented as a nonsquare matrix, since it maps P_{n-1} into P_n, and P_n is not a subspace of P_{n-1}. Thus X will be represented by an $(n+1) \times n$ matrix. We leave it to the reader to show that

$$[X] \equiv (x_{ij}) = \begin{bmatrix} 0 & 0 & 0 & \cdots & & 0 \\ 1 & 0 & 0 & & & 0 \\ 0 & 1 & 0 & & & 0 \\ 0 & 0 & 1 & & & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}. \tag{3.12}$$

We shall return to the matrix representation of the commutator $[D, X]$ after first discussing the addition and multiplication of the matrices of linear transformations.

The problem is to find the matrices associated with the sum and product of two linear transformations whose matrices are known in some fixed basis, $X = \{x_1, \cdots, x_n\}$. Let the matrices of A and B with respect to X be (a_{ij}) and (b_{ij}), respectively. We want to know the matrices of

1. $S \equiv \alpha A + \beta B$,

where α and β are scalars, and

2. $C = AB$,

with respect to the basis X.

1. $$Sx_j = (\alpha A + \beta B)x_j = \alpha Ax_j + \beta Bx_j = \alpha \sum_{i=1}^{n} a_{ij}x_i + \beta \sum_{i=1}^{n} b_{ij}x_i$$
$$= \sum_{i=1}^{n} (\alpha a_{ij} + \beta b_{ij})x_i = \sum_{i=1}^{n} s_{ij}x_i\,,$$

where the final expression simply defines the matrix of S, (s_{ij}). The linear independence of the basis vectors implies that

$$[S]_{ij} = s_{ij} = \alpha a_{ij} + \beta b_{ij} .$$

2. $$Cx_j = ABx_j = A(Bx_j) = A\left(\sum_{k=1}^{n} b_{kj}x_k\right) = \sum_{k=1}^{n} b_{kj}Ax_k$$

$$= \sum_{k=1}^{n} b_{kj} \sum_{i=1}^{n} a_{ik}x_i = \sum_{i,k=1}^{n} a_{ik}b_{kj}x_i = \sum_{i=1}^{n} c_{ij}x_i .$$

Therefore, by the same reasoning as in (1) above

$$[C]_{ij} = c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} . \tag{3.13}$$

This equation defines the multiplication of two $n \times n$ matrices. We derived a special instance of such matrix multiplication in Eq. (1.22).

We can use Eq. (3.13) to define the multiplication of two arrays of numbers. Therefore, the product, AB, of two matrices will be defined if and only if the number of columns of A equals the number of rows of B. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 11 & 17 & 23 \\ 11 & 25 & 39 & 53 \\ 17 & 39 & 61 & 83 \end{bmatrix} .$$

The product of an $(l \times m)$ matrix (l rows and m columns) and an $(m \times n)$ matrix is an $(l \times n)$ matrix. Thus Eq. (3.10) represents the multiplication of an $(n \times n)$ matrix, A, by the $(n \times 1)$ column matrix, α, to yield the $(n \times 1)$ column matrix, β.

Clearly, the matrices of the special linear transformation, O and I, are

$$[O]_{ij} = 0 , \qquad [I]_{ij} = \delta_{ij} . \tag{3.14}$$

Note that for the zero and identity matrices:

$$OA = AO = O , \qquad IA = AI = A ,$$

just as for the corresponding linear transformations.

At this juncture, one could prove that linear transformations and matrices are related isomorphically. The proof is simple and is omitted; the intuitive content of the theorem is only a slight generalization of Theorem 3.5, which demonstrated that vectors are isomorphic to n-tuples. Henceforth we can be sure that any theorem proved for a linear transformation automatically holds for the matrix associated with it relative to a given basis, and vice versa. Thus, the sloppy notation which fails to distinguish between linear transformations and matrices has a measure of theoretical justification.

The most unusual of the properties of linear transformations when contrasted with the properties of numbers is that they need not commute. The matrices of linear transformations will therefore exhibit this same property. From Eq.

(3.13) we see that, in general, $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{jk}$, will not be equal to $(BA)_{ij} = \sum_{k=1}^{n} b_{ik}a_{kj}$.

Example. Consider the products of the matrices of the differentiation and multiplication operators defined in Eqs. (3.11) and (3.12). Since D is an $n \times (n+1)$ matrix and X is an $(n+1) \times n$ matrix, DX will be an $n \times n$ matrix. We have

$$(DX)_{ij} = \sum_{k=1}^{n+1} d_{ik}x_{kj} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 2 & 0 & & 0 \\ 0 & 0 & 3 & & 0 \\ \cdot & & & & \cdot \\ \cdot & & & & \cdot \\ \cdot & & & n-1 & 0 \\ 0 & 0 & 0 \cdots & 0 & n \end{bmatrix}.$$

It is evident from Eqs. (3.11) and (3.12) for D and X that XD will be $(n+1)$-dimensional, whereas DX is n-dimensional. This is sensible, since X maps P_{n-1} into P_n, and D maps P_n back into P_{n-1}; thus DX maps P_{n-1} into P_{n-1}, and DX is n-dimensional. However, XD maps P_n into P_n, because D first maps P_n into P_{n-1} and then X maps P_{n-1} into P_n; therefore XD is $(n+1)$-dimensional. A simple calculation gives

$$(XD)_{ij} = \sum_{k=1}^{n} x_{ik}d_{kj} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & & 0 & 0 \\ 0 & 0 & 2 & & 0 & 0 \\ \vdots & & & & & \vdots \\ 0 & 0 & 0 & & n-1 & 0 \\ 0 & 0 & 0 & & 0 & n \end{bmatrix}.$$

Suppose that we restrict XD to mapping P_{n-1} into P_{n-1}, so that D is $(n-1) \times n$ and X is $n \times (n-1)$. Then we get an $n \times n$ matrix obtained from the above matrix by lopping off the $(n+1)$ row and $(n+1)$ column. If we now compare DX with XD, both operating on P_{n-1}, we see that they do not commute and that in fact

$$DX - XD = I,$$

in agreement with the discussion in Section 3.5.

Even though the matrices D and X are not square for any finite-dimensional space, they may, in a rather imprecise way, be thought of as square infinite-dimensional matrices if n increases without limit, since the difference between n and $n+1$ becomes negligible when n is arbitrarily large. They are often dealt with as such in field theory.

We close this section by introducing a number of special type of matrices.

1. *Diagonal matrices.* The special property of a *diagonal* matrix is that its only nonzero elements lie on its northwest-southeast diagonal, that is, $a_{ij} = 0$ if

$i \neq j$, or $a_{ij} = d_i\delta_{ij}$. Clearly, any two $n \times n$ diagonal matrices commute, and their product is also a diagonal matrix.

2. *Idempotent matrices.* A matrix A is said to be *idempotent* if $A^2 = A$. Thus the matrices I, O, and

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

are idempotent.

3. *The transpose of a matrix.* Given $A = (a_{ij})$, we define the *transpose* of A, written $\tilde{A}$, as

$$[\tilde{A}]_{ij} = a_{ji} .$$

Theorem 3.9.

a) If $\tilde{A} = B$, then $\tilde{B} = A$ (consequently, $\tilde{\tilde{A}} = A$),

b) $\widetilde{A + B} = \tilde{A} + \tilde{B}$,

c) $\widetilde{AB} = \tilde{B}\tilde{A}$.

Proof. The proofs of (a) and (b) are trivial. For (c), let $C = AB$. Then

$$[\tilde{C}]_{ij} \equiv [\widetilde{AB}]_{ij} = [AB]_{ji} = \sum_k [A]_{jk}[B]_{ki} = \sum_k [\tilde{B}]_{ik}[\tilde{A}]_{kj} = [\tilde{B}\tilde{A}]_{ij} ,$$

that is,

$$\tilde{C} \equiv \widetilde{AB} = \tilde{B}\tilde{A} . \qquad \text{QED}$$

4. *The adjoint of a matrix.* We define the adjoint of a matrix A to be the matrix $A^\dagger$ whose elements are given by

$$[A^\dagger]_{ij} = a_{ji}^* .$$

Clearly, this can also be written as $A^\dagger = \tilde{A}^*$. As a simple corollary of Theorem 3.9, we have the following results:

a) If $A^\dagger = B$, then $B^\dagger = A$, that is, $(A^\dagger)^\dagger = A$.

b) $(A + B)^\dagger = A^\dagger + B^\dagger$.

c) $(AB)^\dagger = B^\dagger A^\dagger$.

5. *Symmetric and antisymmetric matrices.* An $n \times n$ matrix A is symmetric if $A = \tilde{A}$, that is, if $a_{ij} = a_{ji}$. A is antisymmetric if $A = -\tilde{A}$; that is, if $a_{ij} = -a_{ji}$. The diagonal elements of an antisymmetric matrix must all be zero. Any matrix A may be expressed as the sum of a symmetric and an antisymmetric matrix, for we may write identically

$$A = \underbrace{\tfrac{1}{2}(A + \tilde{A})}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}(A - \tilde{A})}_{\text{antisymmetric}} ,$$

exactly as for tensors [Eq. (1.97)].

6. *Singular and nonsingular matrices.* An $n \times n$ matrix A is said to be nonsingular if there exists an $n \times n$ matrix B such that $AB = I$. Otherwise A is singular. We shall see that if the matrix of a linear transformation is nonsingular, then the linear transformation is invertible, and conversely; in other words, a necessary and sufficient condition for invertibility, of either A or B, on a *finite*-dimensional vector space is $AB = I$. $BA = I$ is not needed on a finite-dimensional space, as it is in Theorem 3.6.

3.8 DETERMINANTS

Aside from being useful in solving systems of linear algebraic equations, the determinant of a matrix is important because it can give information about the associated linear transformation. For example, we shall prove the general result (anticipated in Section 1.4) that the determinant of a matrix corresponding to an orthogonal transformation is always ± 1. Also, we shall see that the determinant of a matrix A is zero if and only if A is singular.

The determinant of a matrix A, written det A, is a function that acts on a matrix to yield a number.

Definition 3.11. For a given linear transformation, A, which is represented by the $n \times n$ matrix (A_{ij}),

$$\det A \equiv \epsilon_{abc\cdots n} A_{1a} A_{2b} \cdots A_{Nn} , \tag{3.16a}$$

where we are using the Einstein summation convention, as we shall do throughout this section. $\epsilon_{abc\cdots n}$ is the antisymmetric symbol on N objects, defined by

$$\epsilon_{abc\cdots n} \equiv \begin{cases} +1 \text{ if } (abc \cdots n) \text{ is an even permutation of } (1, 2, 3, \cdots, N) , \\ -1 \text{ if } (abc \cdots n) \text{ is an odd permutation of } (1, 2, 3, \cdots, N) , \\ 0, \text{ otherwise.} \end{cases}$$

The reader will note that this notation is not without flaws. The subscripts over which we are summing are denoted by $a, b, c, \cdots$, etc. Clearly, if we have a 100-dimensional vector space, there will not be enough letters in the English alphabet to serve as subscripts in Definition 3.11. However, the reader should have no difficulty in imagining an alphabet with 100 letters which we could use in such an emergency; e.g., English + Greek + Arabic + Hebrew + $\cdots$ + Navaho. In discussing determinants, we shall content ourselves with the familiar alphabet. Thus in this section, when we write A_{Nn}, by n we understand the Nth letter of the alphabet. We could, of course, always resort to using subscripted subscripts, such as $i_1, i_2, \cdots$, etc.; however, since the main problem with determinants is a notational one, it is perhaps useful to pare down the symbols as much as possible.

If $(abc \cdots n)$ is some permutation of $(1, 2, 3, \cdots, N)$ we may write $\epsilon_{abc\cdots n} = (-1)^P$, where P is the number of transpositions of subscripts needed to pro-

duce the normal order $(1, 2, 3, \cdots, N)$. If P is even or odd, we speak of the permutation as being of even or odd parity. This ϵ symbol is the natural generalization of the symbol ϵ_{ijk} which proved so useful in the treatment of vectors and tensors in Chapter 1.

Det A is seen to consist of a sum of $N!$ products, each product consisting of one element from every row and every column. Products containing more than a single term from a given column do not contribute to det A because the corresponding ϵ symbol vanishes. From this it is immediately clear that an equivalent definition for det A is

$$\det A = \epsilon_{abc\cdots n} A_{a1} A_{b2} \cdots A_{nN} . \tag{3.16b}$$

The roles of the rows and columns have simply been reversed, or what is the same thing, the rows have become the columns and vice versa. Thus Eq. (3.16b) tells us that

$$\det A = \det \tilde{A} = (\det A^\dagger)^* . \tag{3.17}$$

This alternative definition of det A means that for any theorem that can be proved for rows, there will be a corresponding theorem that holds for columns.

If at any time in this treatment of the determinants of $n \times n$ matrices the reader cannot see the "forest" (the meaning) for the "trees" (the subscripts), he should simply write out the equations for the 2×2 or 3×3 case. In fact, when any theorem concerning $n \times n$ matrices is stated, it is often very helpful in proving (or disproving!) it to consider its meaning in the 2×2 case. One should not feel that any "loss of face" is involved in trying out these general theorems on the simplest nontrivial special case. After all, the general theorems were originally derived by abstraction from just such special cases.

We now list a number of the properties of determinants.

1. A common factor of each element of a row (or column) may be factored out as a multiplicative constant.
2. If any row (or column) is all zeros, the determinant is zero.
3. $\det I = 1$.
4. Interchanging any two rows (or columns) changes the sign of the determinant.
5. If any two rows (or columns) are equal, the determinant is zero.
6. $\det (AB) = \det A \det B$.
7. A scalar multiple of a row (or column) may be added to another row (or column) without changing the value of the determinant.
8. $\det A = 0$ if and only if the row (or column) vectors of A are linearly dependent.

The proofs of (1) through (3) are simple. We shall prove properties (4), (5), and (6). It will be clear how to prove (7), and then (8), after the proofs of (4), (5), and (6) have been studied; so we shall leave these for the reader to practice on.

The first step in proving (4) is to show that if two *adjacent* rows of a matrix are interchanged, its determinant merely changes sign. Let A' be the matrix obtained from A by interchanging the Kth and Lth rows, which we take to be adjacent. Then

$$\begin{aligned}\det A' &= \epsilon_{a\ldots kl\ldots n} A_{1a} A_{2b} \cdots A_{Lk} A_{Kl} \cdots A_{Nn} \\ &= \epsilon_{a\ldots lk\ldots n} A_{1a} \cdots A_{Kk} A_{Ll} \cdots A_{Nn} = -\det A\,,\end{aligned}$$

where we have interchanged the order of the factors in the product $A_{Kk}A_{Ll}$, interchanged the summed over, dummy subscripts, k and l, and, finally, interchanged the order of lk in the ϵ-symbol which introduces the minus sign. Property (4) now follows if we note that the interchange of *any* two rows can be effected by an odd number of transpositions of adjacent rows. Property (5) is an immediate consequence of (4).

We now derive property (6): The determinant of the product of two matrices is the product of their determinants. We first note that

$$\epsilon_{abc\ldots n} \det A = \epsilon_{a'b'c'\ldots n'} A_{aa'} A_{bb'} \cdots A_{nn'}\,, \tag{3.18a}$$

for if $(abc \cdots n)$ are set equal to $(1, 2, 3, \cdots, N)$, Eq. (3.18a) reduces to the definition of $\det A$ (Eq. 3.16a). If two (or more) of the unprimed subscripts are equal, then both sides of Eq. (3.18a) vanish since the right-hand side represents determinant of a matrix with two (or more) equal rows; and if $(abc \cdots n)$ is a permutation of $(1, 2, 3, \cdots, N)$, then the expressions on the right and left of Eq. (3.18a) will give $(-1)^P \det A$ where P is the parity of the permutation. In exactly the same manner, it follows that

$$\epsilon_{abc\ldots n} \det A = \epsilon_{a'b'c'\ldots n'} A_{a'a} A_{b'b} \cdots A_{n'n}\,. \tag{3.18b}$$

Now consider the product $(\det A)(\det B)$. Applying Eq. (3.16b) to B, we have

$$(\det A)(\det B) = (\det A)\, \epsilon_{abc\ldots n} B_{a1} B_{b2} \cdots B_{nN}\,;$$

using Eq. (3.18b), we may write this as

$$\begin{aligned}(\det A)(\det B) &= \epsilon_{a'b'c'\ldots n'} A_{a'a} A_{b'b} \cdots A_{n'n} B_{a1} B_{b2} \cdots B_{nN} \\ &= \epsilon_{a'b'c'\ldots n'} (A_{a'a} B_{a1})(A_{b'b} B_{b2}) \cdots (A_{n'n} B_{nN}) \\ &= \epsilon_{a'b'c'\ldots n'} (AB)_{a'1} (AB)_{b'2} \cdots (AB)_{n'N} \\ &= \det (AB)\,. \qquad \text{QED}\end{aligned}$$

There is a way to picture this result geometrically that is considerably more intuitive than the proof we have given, and also suggests the relevance of determinants to physics. In three dimensions, the volume of a parallelepiped defined by three vectors **A**, **B**, and **C** [Fig. (3.3)] is

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \epsilon_{ijk} A_i B_j C_k = \det \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix}.$$

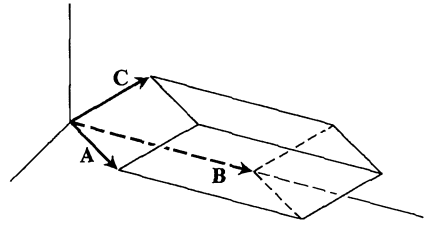

Fig. 3.3 Parallelepiped defined by three vectors, **A**, **B**, and **C**.

Given this interpretation of the quantity $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})$ as a *volume,* it is not surprising that it is invariant under rotations of the coordinate system, and hence is a scalar. Actually, it is a pseudoscalar; it changes sign under inversion of the coordinate system. Hence the physical volume is really $|\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})|$.

This association of a determinant with a volume may be generalized to more or fewer dimensions. In two dimensions, the "volume" becomes the area of a parallelogram determined by two vectors in a plane. Let I denote the matrix which corresponds to the unit square:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The row vectors of I determine the endpoints of the sides of the unit square with one corner at the origin. Now a matrix T multiplying I and giving T may be viewed as transforming the unit square to a parallelogram defined by the row vectors of T. The area of this parallelogram is det T.

We can decompose an arbitrary parallelogram represented by a matrix A, of area det A, into little *squares* of arbitrarily small area s_i where $\sum_i s_i =$ det A. Then a matrix T multiplying A, and changing it into another parallelogram whose matrix is TA, can be viewed as operating on each of the little squares of which A consists. Under the transformation, each little of area s_i becomes a little parallelogram of area $a_i = (\det T)s_i$. Therefore the new parallelogram whose matrix is TA has an area of

$$\sum_i a_i = \sum_i (\det T)s_i = (\det T)\left(\sum_i s_i\right) = \det T \det A\,.$$

But the area of the parallelogram represented by the matrix TA is also given by det (TA). Therefore det $TA =$ det T det A.

The key ideas of this proof expressed in three-dimensional language are:

1. The unit cube is carried by a linear transformation T into a parallelepiped of volume det T. The determinant of the linear transformation gives the ratio of the final volume to that of the original cube.

2. The determinant of the matrix of a linear transformation gives the ratio of the volumes of two arbitrary parallelepipeds whose corresponding matrices are connected by the matrix of the linear transformation. The determinant is thus a scale factor for measuring volume changes.

3. If we go from one volume to another to a third, and so on, we accumulate these scale factors, and the total factor relating the initial and the final volume is the product of all the determinants of all the matrices of the linear transformations which relate the different parallelepipeds to each other. On the other hand, the first parallelepiped is related to the last by the product of all the matrices; the volume of the parallelepiped that results from the successive transformations is just the determinant of this product of matrices. Thus the product of the determinants of n matrices equals the determinant of the product of the n matrices. This proof holds for spaces of any dimension.

A very important combinatorial result that reduces the evaluation of the determinant of an $n \times n$ matrix to the evaluation of the determinants of $(n-1) \times (n-1)$ matrices is the following:

Theorem 3.10. $\det A = \sum_j A_{Jj} \operatorname{cof}(A_{Jj})$, for any row J, that is, J is not summed on. We include the summation on j explicitly to remind the reader that J is *not* summed. By $\operatorname{cof}(A_{Jj})$ we mean the cofactor of the element A_{Jj}, which equals $(-1)^{J+j}$ times the determinant of the $(n-1) \times (n-1)$ matrix obtained from A by striking out the Jth row and the jth column.

Proof.
$$\begin{aligned}\det A &= \epsilon_{abc\ldots n} A_{1a} A_{2b} \cdots A_{Nn} \\ &= \sum_j A_{Jj}[\epsilon_{a\ldots ijk\ldots n} A_{1a} \cdots A_{Ii} A_{Kk} \cdots A_{Nn}] ,\end{aligned} \tag{3.19}$$

where we have put the cofactor of A_{Jj} in brackets. Now

$$\operatorname{cof}(A_{Jj}) = (-1)^{J-1} \epsilon_{ja\ldots ik\ldots n} A_{1a} \cdots A_{Ii} A_{Kk} \cdots A_{Nn} , \tag{3.20}$$

where we have moved the subscript j in the symbol ahead of the others by $(J-1)$ transpositions. (Remember that j means the Jth letter of the alphabet!) These transpositions produce a factor $(-1)^{J-1}$. Now if the reader will stare at Eq. (3.20) for a moment, he will see that $\operatorname{cof}(A_{Jj})$ is nearly equal to what we asserted in the statement of the theorem: we have removed the Jth row, so there is no factor A_{Jj}, and in summing on the set $(a \cdots ik \cdots n)$ no subscript can take on the numerical value of j (remember that in Eq. (3.20) j is fixed at some integer between 1 and N), because we have excluded the jth column. Only the j in the ϵ-symbol stands between us and our result. But

$$\epsilon_{ja\ldots ik\ldots n} = (-1)^{j-1} \tilde{\epsilon}_{a\ldots ik\ldots n} , \tag{3.21}$$

where on the right-hand side of Eq. (3.21) the subscripts $a, \cdots, i, k, \cdots, n$ take on all values between 1 and N, *excluding* the (fixed) value taken by j. The normal ordering in this $\tilde{\epsilon}$ symbol will be the usual chronological order, $1, 2, \cdots, j-1, j+1, \cdots, N$. Thus we obtain

$$\operatorname{cof}(A_{Jj}) = (-1)^{J+j} \tilde{\epsilon}_{a\ldots ik\ldots n} A_{1a} \cdots A_{Ii} A_{Kk} \cdots A_{Nn} ,$$

where in the $N-1$ summations implied by the Einstein convention, the indices *may not take on the value* j. Therefore the cofactor is just $(-1)^{J+j}$ times the

determinant of the $(n-1) \times (n-1)$ matrix obtained from A by striking out the Jth row and the jth column. QED

An analog to Theorem 3.10 for columns is obtained in exactly the same way. It is

$$\det A = \sum_j A_{jJ} \operatorname{cof}(A_{jJ}) , \tag{3.22}$$

for any column J, that is, no summation on J.

A simple and useful generalization of this basic result is

Theorem 3.11.

$$\delta_{KJ} \det A = \sum_j A_{Kj} \operatorname{cof}(A_{Jj}) , \tag{3.23a}$$

for all J and K.

Proof. For $J = K$, this reduces to Theorem 3.10. For $J \neq K$, the left-hand side vanishes as does the right-hand side since the expansion on the right-hand side is equivalent to the expression for the determinant of a matrix with two equal rows. To see this, we write

$$\begin{aligned}\sum_j A_{Kj} \operatorname{cof}(A_{Jj}) &= \sum_j A_{Kj}(-1)^{J+j} \tilde{\epsilon}_{a\ldots ik\ldots n} A_{1a} \cdots A_{Ii} A_{Kk} \cdots A_{Nn} , \\ &= \sum_j \epsilon_{ja\ldots ik\ldots n} A_{1a} \cdots A_{Ii} A_{Kj} A_{Kk} \cdots A_{Nn} (-1)^{J-1} \\ &= \sum_j \epsilon_{a\ldots ijk\ldots n} A_{1a} \cdots A_{Ii} A_{Kj} A_{Kk} \cdots A_{Nn} .\end{aligned}$$

This last expression is just the determinant of a matrix containing two equal rows (the Kth) and hence vanishes. QED

Here we have just reversed the reasoning which led in Theorem 3.10 to the expression for cof (A_{Jj}). Similarly, it follows that

$$\delta_{JK} \det A = \sum_j A_{jK} \operatorname{cof}(A_{jJ}) , \tag{3.23b}$$

for all J and K.

Before ending our discussion of determinants, we should perhaps say a final word about the $\tilde{\epsilon}$ symbol we defined in proving Theorem 3.10. Any ϵ-type symbol is just a way of ordering a set of integers. The ϵ-symbol orders the first N integers, while $\tilde{\epsilon}$ orders the first N integers with *one* integer missing, i.e., it has only $N-1$ subscripts (if it is to be nontrivial). Clearly, we could also define $\tilde{\tilde{\epsilon}}$, which would order the first N integers with *two* integers missing. In all such cases the defining properties of the symbol are completely analogous to those which we listed for the ϵ-symbol. The relation, Eq. (3.21),

$$\epsilon_{ja\ldots ik\ldots n} = (-1)^{j-1} \tilde{\epsilon}_{a\ldots ik\ldots n}$$

is evident when one recognizes that the same number of transpositions is required to order $(a \cdots ik \cdots n)$ numerically in ϵ as in $\tilde{\epsilon}$. Then an additional $j-1$

transpositions are required to get j to the correct position in ϵ. Hence the factor $(-1)^{j-1}$ multiplying $\tilde{\epsilon}$.

We now proceed to apply these results to the theory of linear operators.

Definition 3.11. The *classical adjoint* of an $n \times n$ matrix A is the $n \times n$ matrix, adj A, where $(\text{adj } A)_{ij} = \text{cof } (A_{ji})$.

It follows directly from Eqs. (3.23a) and (3.23b) that

$$A(\text{adj } A) = (\text{adj } A)A = (\det A)I \,. \tag{3.24}$$

This is a matrix equation. The matrix $[(\det A)I]$ is diagonal, with $\det A$ for every diagonal element. (A word of caution; the *classical adjoint* is not the same as the *adjoint operator* to be defined in Section 4.4). Now we prove an extremely useful theorem.

Theorem 3.12. The matrix A has an inverse, A^{-1}, if and only if $\det A \neq 0$.

Proof. If $\det A \neq 0$, we may divide Eq. (3.24) by $\det A$ to get

$$A\left(\frac{\text{adj } A}{\det A}\right) = \left(\frac{\text{adj } A}{\det A}\right)A = I \,.$$

By theorem 3.6, this implies that there exists a unique inverse

$$A^{-1} = \left(\frac{\text{adj } A}{\det A}\right) .$$

Conversely, if there exists an inverse, A^{-1}, then

$$AA^{-1} = I = A^{-1}A \,.$$

Now take the determinant of both sides to get

$$\det (AA^{-1}) = \det A \cdot \det A^{-1} = \det I = 1 \,, \tag{3.25}$$

which implies that $\det A \neq 0$. QED

Corollary. A matrix A is nonsingular if and only if $\det A \neq 0$.

Proof. If A is nonsingular, then by definition there exists a matrix B such that $AB = I$. It follows that $\det A \det B = 1$, so that $\det A \neq 0$. If $\det A \neq 0$, then from theorem (3.12) A has a unique inverse, $A^{-1} = \text{adj } A/\det A$, so there exists a matrix $B = A^{-1}$ such that $AB = I$. Thus, as mentioned previously, having an inverse is the same thing as being nonsingular (in a finite-dimensional vector space). Some authors define a matrix A as nonsingular if $\det A \neq 0$ (and singular if $\det A = 0$), and then follow a slightly different logical route to what is the same key result: that a nonsingular matrix has a unique inverse given by

$$A^{-1} = \frac{1}{\det A} (\text{adj } A) \,. \tag{3.26}$$

It follows from Eq. (3.25) that

$$\det(A^{-1}) = \frac{1}{\det A}\,. \tag{3.27}$$

The expression $1/A$ is meaningless because A is a matrix; "$1/banana$" is equally meaningless. But $1/\det A \equiv [\det A]^{-1}$ is a scalar because $\det A$ is a scalar-valued function of a matrix A.

Once we have a way of computing the inverse of a matrix, we can solve systems of linear equations, such as $Ax = y$. In the three-dimensional case this shorthand notation [introduced in Eqs. (3.9) and (3.10)] stands for

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{bmatrix}.$$

By carrying out the indicated multiplication, we get the system of three equations in three unknowns:

$$\begin{aligned} a_{11}\xi_1 + a_{12}\xi_2 + a_{13}\xi_3 &= \eta_1 \\ a_{21}\xi_1 + a_{22}\xi_2 + a_{23}\xi_3 &= \eta_2 \\ a_{31}\xi_1 + a_{32}\xi_2 + a_{33}\xi_3 &= \eta_3\,. \end{aligned} \tag{3.28}$$

Now if $\det A \neq 0$, A^{-1} exists and is given by

$$A^{-1} = [\det A]^{-1} \operatorname{adj} A\,.$$

Then

$$A^{-1}Ax = Ix = A^{-1}y = [\det A]^{-1} (\operatorname{adj} A)\, y\,. \tag{3.29}$$

This solves the system for the unknown ξ's in terms of the given η's. Writing out Eq. (3.29) explicitly gives

$$\xi_i = (\det A)^{-1} \sum_{j=1}^{n} [\operatorname{adj} A]_{ij}\eta_j = (\det A)^{-1} \sum_{j=1}^{n} \operatorname{cof}(A_{ji})\eta_j\,. \tag{3.30}$$

The quantity

$$\sum_{j=1}^{n} \eta_j \operatorname{cof}(A_{ji})$$

is just the determinant of the matrix obtained by replacing the ith column of A by the η's. Equation (3.30) is known as *Cramer's rule*. It is usually *not* the most efficient way to solve large linear systems; there are numerical techniques that are much faster. These are discussed in books on numerical analysis.

For a homogeneous system of equations, where η_j vanishes for all j, Cramer's rule implies that we can only have a nontrivial solution (i.e., some ξ's $\neq 0$) if $\det A = 0$. We can also see this directly from Theorem 3.7, which says that if A is nonsingular, then $Ax = 0$ implies that $x = 0$. Clearly, then,

a necessary condition for a nontrivial solution ($x \neq 0$) will be the singularity of A. It is shown in books on linear equations that this is also a sufficient condition for a nontrivial solution. However, the reader should *not* deduce that if $\det A = 0$, then the equation $Ax = y$ has no solution unless $y = 0$. This is false, as Theorem 3.17 will show.

Example. We calculate the inverse of the matrix

$$A = (a_{ij}) = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 0 \\ 1 & 2 & 3 \end{bmatrix}.$$

First, we note that $\det A = -3$, so that A is nonsingular and has an inverse. From Eq. (3.26), we find that

$$[A^{-1}]_{ij} = \frac{1}{\det A}[\operatorname{adj} A]_{ij} = \frac{1}{\det A}\operatorname{cof}(A_{ji});$$

$$A^{-1} = -\frac{1}{3}\begin{bmatrix} (-1)^{1+1}\det\begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix} & (-1)^{2+1}\det\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} & (-1)^{3+1}\det\begin{pmatrix} 3 & 4 \\ 1 & 0 \end{pmatrix} \\ (-1)^{1+2}\det\begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} & (-1)^{2+2}\det\begin{pmatrix} 1 & 4 \\ 1 & 3 \end{pmatrix} & (-1)^{3+2}\det\begin{pmatrix} 1 & 4 \\ 2 & 0 \end{pmatrix} \\ (-1)^{1+3}\det\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} & (-1)^{2+3}\det\begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix} & (-1)^{3+3}\det\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} \end{bmatrix}$$

$$= -\frac{1}{3}\begin{bmatrix} 3 & -1 & -4 \\ -6 & -1 & 8 \\ 3 & 1 & -5 \end{bmatrix} = \begin{bmatrix} -1 & \frac{1}{3} & \frac{4}{3} \\ 2 & \frac{1}{3} & -\frac{8}{3} \\ -1 & -\frac{1}{3} & \frac{5}{3} \end{bmatrix}.$$

The reader may check the result by showing that

$$AA^{-1} = A^{-1}A = I.$$

Again it should be understood that there are other numerical methods for inverting matrices which are far more efficient for many purposes.

3.9 SIMILARITY TRANSFORMATIONS

We now investigate the changes induced in the matrix representative of a linear transformation when the set of basis vectors with respect to which it is expressed is changed. Let

$$X = \{x_1, \cdots, x_n\}$$

and

$$Y = \{y_1, \cdots, y_n\}$$

be two given bases in an n-dimensional vector space. We assume that they are related to each other by the linear transformation A:

$$y_i = Ax_i.$$

Now any linear transformation that transforms one basis into another must be invertible, because for any vector

$$x = \sum_i \alpha_i x_i ,$$

if $Ax = 0$, then

$$A\left(\sum_i \alpha_i x_i\right) = \sum_i \alpha_i A x_i = \sum_i \alpha_i y_i = 0 ,$$

which implies that $\alpha_i = 0$, since the y_i are linearly independent; thus $x = 0$. Therefore, by theorem 3.7, A^{-1} exists.

Suppose that we expand a given vector x in the two bases X and Y:

$$x = \sum_i \alpha_i x_i \qquad \text{and} \qquad x = \sum_i \beta_i y_i .$$

Now the question is, how are α_i and β_i related? Or, to put it another way, how do the coordinates transform under a change of basis? We have

$$x = \sum_i \beta_i y_i = \sum_i \beta_i A x_i = \sum_i \beta_i \sum_j a_{ji} x_j = \sum_{i,j} a_{ji} \beta_i x_j .$$

But

$$x = \sum_j \alpha_j x_j ;$$

therefore, with a change of dummy subscripts, we get

$$\sum (\alpha_j - \sum_i a_{ji}\beta_i) x_j = 0 \Longrightarrow \alpha_i = \sum_j a_{ij}\beta_j .$$

This equation tells us how the coordinates of a given vector change when we go from one basis to another. It is exactly of the form of Eq. (1.15b), in which the same question is answered, under the same convention for defining matrix elements; there the x_i correspond to the present α_i. We may write it in the abbreviated form

$$\alpha_X = [A]_X \beta_Y ,$$

where the subscripts serve to remind us that the column vectors α and β represent x in the X and Y bases, respectively, and that the matrix representative of A is in the X basis.

We now turn to the analogous question for the matrix representation of a linear transformation, B. What is the relation between the matrix representation (b_{ij}) of B with respect to the basis X, and its representation with respect to the basis Y, (b'_{ij})? We have

$$Bx_j = \sum_i b_{ij} x_i \qquad \text{and} \qquad By_j = \sum_i b'_{ij} y_i .$$

We know that

$$By_j = BAx_j = B \sum_k a_{kj} x_k ,$$

where (a_{kj}) is the matrix representation of A with respect to X. Thus

$$By_j = \sum_k a_{kj}Bx_k = \sum_k a_{kj} \sum_i b_{ik}x_i = \sum_i (\sum_k b_{ik}a_{kj})x_i \,. \tag{3.31}$$

But we also have

$$By_j = \sum_k b'_{kj}y_k = \sum_k b'_{kj}Ax_k = \sum_k b'_{kj} \sum_i a_{ik}x_i = \sum_i (\sum_k a_{ik}b'_{kj})x_i \,. \tag{3.32}$$

The linear independence of the basis vectors x_i now allows us to deduce from Eqs. (3.31) and (3.32) that

$$\sum_k b_{ik}a_{kj} = \sum_k a_{ik}b'_{kj} \,, \tag{3.33}$$

or, in matrix notation,

$$[B]_X[A]_X = [A]_X[B]_Y \,. \tag{3.34}$$

The subscripts indicate the basis with respect to which the matrix representative is determined. Since A has an inverse, we may write Eq. (3.34) as

$$[B]_Y = [A]_X^{-1}[B]_X[A]_X \,. \tag{3.35}$$

Definition 3.12. Two matrices, B and C, are said to be *similar* if there exists an invertible matrix, A, such that $C = A^{-1}BA$; we say that C is related to B by a *similarity transformation*.

Thus the previous result may be stated as "similar matrices represent the same linear transformation with respect to different bases."

Now we prove two important theorems about similar matrices.

Theorem 3.13. If B and C are similar matrices,

$$\det B = \det C \,.$$

Proof.

$$\det C = \det (A^{-1}BA) = \det A^{-1} \det B \det A = \det B \,,$$

from Eq. (3.27). QED

Definition 3.13. The *trace* of a matrix A, written $\operatorname{tr} A$, is the sum of the diagonal elements:

$$\operatorname{tr} A \equiv \sum_i a_{ii} \,.$$

Theorem 3.14. If B and C are similar matrices, $\operatorname{tr} B = \operatorname{tr} C$.

Proof.

$$\begin{aligned}\operatorname{tr} C &= \sum_i c_{ii} = \sum_i [A^{-1}BA]_{ii} = \sum_{i,j,k} [A^{-1}]_{ij}[B]_{jk}[A]_{ki} \\ &= \sum_{i,j,k} [A]_{ki}[A^{-1}]_{ij}[B]_{jk} = \sum_{j,k} [AA^{-1}]_{kj}[B]_{jk} \\ &= \sum_{j,k} \delta_{kj}[B]_{jk} = \sum_k [B]_{kk} = \operatorname{tr} B \,. \qquad \text{QED}\end{aligned}$$

Therefore both the determinant and the trace are invariant under similarity transformations. The reader may prove as a simple exercise that tr $(AB) =$ tr (BA) for arbitrary $n \times n$ matrices A, B. Theorem (3.14) may be derived without recourse to summations and subscripts if one uses this result.

3.10 EIGENVALUES AND EIGENVECTORS

Definition 3.14. A scalar λ is an *eigenvalue* and a *nonzero* vector x is an *eigenvector* of a linear transformation A if

$$Ax = \lambda x .$$

When λ and x are eigenvalues and eigenvectors of A, all that A does in operating on x is to multiply x by a constant, thus possibly changing its length, but not changing its "direction."

In quantum mechanics, it is assumed that any observable that exists in nature can be represented by a linear operator A and that the result of any physical measurement of the observable must be one of the eigenvalues of A. For example, the allowed values of the energy of a system with a Hamiltonian H are the values of E which satisfy $Hu = Eu$, where u denotes the eigenvectors describing the physical states associated with the allowed energies E. This is Schrödinger's time-independent equation.

We now investigate the problem of determining the eigenvalues and eigenvectors of an operator. The matrix equation $Ax = \lambda x$ is the same as $Ax - \lambda Ix = 0$ or $(A - \lambda I)x = 0$. We know that if $(A - \lambda I)$ is nonsingular, then this equation has only the trivial solution, $x = 0$. So in order for there to be an eigenvector, the matrix $(A - \lambda I)$ must be singular. That is,

$$\det (A - \lambda I) = \det \begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{bmatrix} = 0 . \qquad (3.36)$$

Equation (3.36) is called the *characteristic equation* for A. The polynomial $\det (A - \lambda I)$ is called the *characteristic polynomial*. The eigenvalues of A, which are usually labeled λ_i, are the n roots (not necessarily distinct) of the characteristic polynomial. The set of eigenvalues $\{\lambda_i\}$ is called the *spectrum* of A. If a given eigenvalue is a multiple root of the characteristic polynomial, the spectrum is said to be *degenerate*.

We know from algebra that every polynomial of degree n has n (not necessarily distinct) roots in the complex field. In fact, we shall prove this fundamental result in Chapter 6. Of course, it is not true that every polynomial has a root in the *real* field (e.g., the polynomial $x^2 + 1$ has only the roots $\pm i$). In order to ensure that the operators we deal with have eigenvalues and eigenvectors, we shall henceforth work in the complex field, unless we explicitly restrict the discussion to the real field.

By putting the eigenvalues back into the equation $Ax_i = \lambda_i x_i$ one at a time, we can solve for the eigenvector x_i that belongs to each λ_i. Note that any linear combination of eigenvectors belonging to the same eigenvalue is itself an eigenvector belonging to that eigenvalue. If we let x_i and x_j be eigenvectors belonging to the same eigenvalue λ, then

$$Ax_i = \lambda x_i \qquad \text{and} \qquad Ax_j = \lambda x_j ,$$

and

$$A(\alpha x_i + \beta x_j) = \alpha A x_i + \beta A x_j = \alpha\lambda x_i + \beta\lambda x_j = \lambda(\alpha x_i + \beta x_j) .$$

Definition 3.15. The *geometric multiplicity* of an eigenvalue λ is the number of linearly independent eigenvectors belonging to λ.

Another type of multiplicity, the algebraic multiplicity, may also be defined. The *algebraic multiplicity* of a given root is the number of times it is repeated in the solution to the characteristic equation. It must be carefully distinguished from the notion of geometric multiplicity defined here.

Theorem 3.15. If A and B are similar, then A and B have the same eigenvalues and these eigenvalues have the same geometric multiplicities.

Proof. Consider $Ax_i = \lambda_i x_i$ and suppose that $P^{-1}AP = B$. Then

$$P^{-1}Ax_i = \lambda_i P^{-1}x_i ,$$

so

$$P^{-1}APP^{-1}x_i = \lambda_i P^{-1}x_i ,$$

where we have simply inserted $I = PP^{-1}$ into the left-hand side of the equation. Thus

$$By_i = \lambda_i y_i , \qquad \text{where} \qquad y_i = P^{-1}x_i ,$$

and it follows that B has the same eigenvalues, λ_i, as A.

The eigenvector of B belonging to λ_i is $y_i = P^{-1}x_i$, and we must prove that if there are k linearly independent eigenvectors x_i satisfying $Ax_i = \bar{\lambda}x_i$—so that the geometric multiplicitly of $\bar{\lambda}$ is k—then there are k linearly independent eigenvectors y_i satisfying $By_i = \bar{\lambda}y_i$. Consider a linear combination of the eigenvectors y_i belonging to $\bar{\lambda}$:

$$\sum_{i=1}^{k} \alpha_i y_i = \sum_{i=1}^{k} \alpha_i P^{-1}x_i = P^{-1}\Big(\sum_{i=1}^{k} \alpha_i x_i\Big) .$$

Thus if we assume

$$\sum_{i=1}^{k} \alpha_i y_i = 0 ,$$

it follows that

$$P^{-1}\Big(\sum_{i=1}^{k} \alpha_i x_i\Big) = 0 .$$

Since P exists, this implies that

$$\sum_{i=1}^{k} \alpha_i x_i = 0 .$$

Hence $\alpha_i = 0$, by the assumed linear independence of the k eigenvectors x_i. Therefore

$$\sum_{i=1}^{k} \alpha_i y_i = 0 \Longrightarrow \alpha_i = 0 ,$$

so the k eigenvectors, y_i, must also be linearly independent. QED

A somewhat cryptic alternative proof that similar matrices have the same eigenvalues consists of the observation that

$$(B - \lambda I) = P^{-1}AP - \lambda I = P^{-1}(A - \lambda I)P ,$$

and therefore

$$\det (B - \lambda I) = \det (A - \lambda I) .$$

Thus, since the two similar matrices A and B have the same characteristic polynomial, they must have the same eigenvalues, and these eigenvalues must have the same *algebraic* multiplicities. Note that this result is *not* equivalent to that of Theorem 3.15. The fact that similar matrices have the same characteristic polynomial implies that the coefficients of each power of λ must be equal. Examination of the characteristic polynomial will show that the coefficient of λ^0 (the constant term in the polynomial) is $\det A$; thus $\det A = \det B$. And since the coefficient of λ^{n-1} is $\operatorname{tr} A$, it follows that $\operatorname{tr} A = \operatorname{tr} B$. This is, in fact, an alternative proof of the invariance of the determinant and trace under similarity transformations; indeed, it points to the existence of a whole class of such invariants, namely, the n coefficients of the various powers of λ ($\lambda^0, \lambda^1, \lambda^2, \cdots, \lambda^{n-1}$) in the characteristic polynomial of an nth order matrix. We shall soon see why the determinant and trace are singled out as having special significance.

If A were a diagonal matrix of the form

$$D = \begin{bmatrix} d_{11} & 0 & 0 & \cdots & 0 \\ 0 & d_{22} & 0 & & 0 \\ 0 & 0 & d_{33} & & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \cdots & d_{nn} \end{bmatrix} ,$$

then the equation $Dx = \lambda x$ would be, in effect, already solved. For in this case the characteristic equation is

$$\det (D - \lambda I) = \prod_{i=1}^{n} (d_{ii} - \lambda) = 0 ,$$

which has the solutions

$$\lambda_i = d_{ii} , \qquad i = 1, 2, \cdots, n ;$$

that is, the eigenvalues are simply the diagonal elements. Thus if we can find a similarity transformation that diagonalizes a matrix A (and as we have seen, such transformations do preserve eigenvalues), we shall have simultaneously solved for the eigenvalues of A.

An important result in this connection is

Theorem 3.16. Let $X_i = (p_{1i}, \cdots, p_{ni})$ be an eigenvector of A (defined on V_n) belonging to λ_i, $(i = 1, \cdots, n)$, so that $AX_i = \lambda_i X_i$. If the vectors X_i span V_n, then the matrix $P = (p_{ij})$ is a diagonalizing matrix for A:

$$P^{-1}AP = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} = D = (\lambda_i \delta_{ij}) = (d_{ij}) .$$

Conversely, if A can be diagonalized by a similarity transformation, then the set of eigenvectors of A spans V.

Proof. The *column* vectors of P are the eigenvectors, X_i, of A. If these n vectors span V, they are linearly independent, so $\det P \neq 0$ and P^{-1} exists.

We have

$$AX_j = \lambda_j X_j ,$$

or

$$\sum_k a_{ik} p_{kj} = \lambda_j p_{ij} = \sum_k p_{ik} \delta_{kj} \lambda_j = \sum_k p_{ik} d_{kj} ,$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n$. Thus

$$AP = PD \Longrightarrow D = P^{-1}AP ,$$

since P is invertible. The converse follows by simply reversing the steps. QED

Thus a spanning set of eigenvectors is both a necessary and sufficient condition for diagonalizability by a similarity transformation. An example (see Problem 18) of a matrix whose eigenvectors do not span the space is

$$A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} .$$

Note that one can always find a P such that $AP = PD$ even if there is no P such that $P^{-1}AP = D$.

Since any matrix A has at least one eigenvalue (call it λ) and one eigenvector (call it p), then the matrix P, built with n columns, each of which is p, will satisfy $AP = PD$. In fact, $AP = \lambda P$, so $D = \lambda I$. But this P is not invertible, since its columns are equal.

Theorem 3.16 is not really very useful in computations because one can only construct the diagonalizing matrix P if one knows the eigenvectors, and usually one can find the eigenvectors only after one knows the eigenvalues. Thus to get the eigenvalues of A by constructing a matrix P that diagonalizes

A, one must already know the eigenvalues of A. The main significance of this theorem is that it provides a necessary and sufficient condition for diagonalizability.

Example 1. We shall compute the eigenvalues, eigenvectors, and a diagonalizing matrix P for the Pauli spin matrix

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

In quantum mechanics the operator for the x-component of spin is

$$s_x = \frac{\hbar}{2}\sigma_x .$$

The characteristic equation for σ_x is

$$\det(\sigma_x - \lambda I) = [(-\lambda)^2 - 1] = 0 \Longrightarrow \lambda = \pm 1 ;$$

thus the spin, s_x, has the eigenvalues $\pm\hbar/2$. To determine the eigenvectors that correspond to these eigenvalues, we put them back one at a time, into the equation $\sigma_x x = \lambda x$. For $\lambda = +1$,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} \Longrightarrow \xi_1 = \xi_2 = a ,$$

so an eigenvector belonging to $\lambda = +1$ has the form

$$\begin{bmatrix} a \\ a \end{bmatrix}, \qquad a \neq 0 .$$

Similarly, an eigenvector belonging to $\lambda = -1$ has the form

$$\begin{bmatrix} b \\ -b \end{bmatrix}, \qquad b \neq 0 .$$

Thus

$$P = \begin{bmatrix} a & b \\ a & -b \end{bmatrix}, \qquad \text{and} \qquad \det P = -2ab \neq 0 .$$

Therefore P^{-1} exists; it is given explicitly by

$$P^{-1} = \frac{1}{\det P}(\operatorname{adj} P) = \frac{-1}{2ab}\begin{bmatrix} -b & -b \\ -a & a \end{bmatrix}.$$

Now

$$P^{-1}\sigma_x P = \frac{1}{2ab}\begin{bmatrix} b & b \\ a & -a \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ a & -b \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$$

Another way to solve for the eigenvalues in this particular case is to note that since the square of any Pauli spin matrix is the identity matrix (Problem 12), that, in particular

$$x = Ix = \sigma_x^2 x = \sigma_x\sigma_x x = \sigma_x\lambda x = \lambda\sigma_x x = \lambda^2 x \Longrightarrow \lambda^2 = 1 ;$$

since $x \neq 0$, we see that $\lambda = \pm 1$. This calculation shows that the eigenvalues of any matrix whose square is I are ± 1.

Because equations of the type $Ax = 0$ are so important in physics, one often gets the impression that since $Ax = 0$ has a solution only if $\det A = 0$, then if $\det A = 0$, an equation of the type $Ax = b$ has no solution unless $b = 0$. This is false. Clearly, $Ax = b$ cannot have a solution for all vectors b; there is, nevertheless, a large class of vectors for which a solution exists. In fact, we can prove the following result, which will be useful when we develop the perturbation theory in Chapter 4.

Theorem 3.17. If $\det A = 0$, and if the eigenvectors of A span the space V on which A acts, then $Ax = b$ has a solution if and only if b may be written as a linear combination of those eigenvectors of A which belong to nonzero eigenvalues. We shall sometimes speak of such a vector b as *lying in the subspace* of V spanned by those eigenvectors of A which belong to nonzero eigenvalues.

Proof. We begin by noting that the determinant of A is equal to the product of the eigenvalues of A (see Problem 3.19). Since $\det A = 0$, it follows that A must have *at least* one zero eigenvalue. We label the eigenvalues by $\lambda_1, \lambda_2, \cdots, \lambda_n$ and the corresponding eigenvectors by $x_1, x_2, \cdots, x_n$. We will include degenerate eigenvalues in our labeling as many times as they occur and make the labeling convention that the zero eigenvalues come first, i.e., if we have m eigenvalues equal to zero, then $\lambda_1 = 0, \lambda_2 = 0, \cdots, \lambda_m = 0$. Since the x_i span the space, we may write

$$b = \sum_{i=1}^{n} \beta_i x_i .$$

Similarly,

$$x = \sum_{i=1}^{n} \xi_i x_i ,$$

so the equation $Ax = b$ may be written

$$A \sum_{i=1}^{n} \xi_i x_i = \sum_{i=1}^{n} \beta_i x_i \Longrightarrow \sum_{i=1}^{n} (\lambda_i \xi_i - \beta_i) x_i = 0 .$$

Since the x_i span V, we have $\lambda_i \xi_i - \beta_i = 0$, for all i. Thus, if a solution exists, then $\beta_i = 0$ for $i = 1, 2, \cdots, m$, since $\lambda_i = 0$ for $i = 1, 2, \cdots, m$. This means that if a solution exists, b may be written as a linear combination of just those x_i belonging to nonzero eigenvalues, and so half the theorem is proved.

On the other hand, if $\beta_i = 0$ for $i = 1, 2, \cdots, m$, then we can solve for the remaining ξ_i:

$$\xi_i = \beta_i / \lambda_i , \qquad i = m + 1, \cdots, n ,$$

while the first m of the ξ_i are arbitrary. Thus any vector of the form

$$x = \xi_1 x_1 + \xi_2 x_2 + \cdots + \xi_m x_m + \frac{\beta_{m+1}}{\lambda_{m+1}} x_{m+1} + \cdots + \frac{\beta_n}{\lambda_n} x_n$$

is a solution to $Ax = b$, so we have an m-fold infinite family of solutions for any b obeying the conditions of the theorem! QED

This arbitrariness in x should not be surprising. Given a solution vector x and an arbitrary multiple of any solution of $Ay = 0$, the sum $x + ay$ is a solution. For if $Ax = b$, then

$$A(x + ay) = Ax + aAy = Ax = b\,.$$

But if A has m eigenvalues equal to zero and m associated linearly independent eigenvectors, $x_1, x_2, \cdots, x_m$, then of course

$$Ax_i = 0x_i = 0\,, \qquad i = 1, \cdots, m\,,$$

so the eigenvectors $x_1, x_2, \cdots, x_m$ are solutions of the homogeneous equation $Ay = 0$. Thus their coefficients in the expansion of x will be arbitrary. This also illustrates the fact that an equation of the type $Ay = 0$ may have more than one linearly independent solution.

The set of all vectors y satisfying $Ay = 0$ is called the *null space* or *kernel* of A; the dimension of this space is called the *nullity* of A. The dimension of the space V on which A acts, minus the nullity of A, is called the *rank* of A.

We now turn to a second example of the eigenvalue-eigenvector problem which is of paramount importance in modern physics.

Example 2. *Energy Eigenvalues and Eigenvectors of the Simple Harmonic Oscillator by Operator Algebra.* The eigenvalue-eigenvector problem for the one-dimensional simple harmonic oscillator is to find the allowed energies, E_n, and the corresponding eigenstates for the quantum-mechanical system whose Hamiltonian is

$$H = (p^2/2m) + \tfrac{1}{2}m\omega^2x^2\,. \tag{3.37}$$

The spring constant k has been replaced using the relation $\omega = (k/m)^{1/2}$. We must solve the equation

$$Hu_n = E_nu_n\,, \tag{3.38}$$

called Schrödinger's equation, for the energy eigenvalues E_n, and the corresponding eigenvectors u_n.

One way to solve this eigenvalue-eigenvector problem is to make the usual Schrödinger substitutions

$$p \to -i\hbar\frac{\partial}{\partial x}\,, \qquad x \to x\,, \tag{3.39}$$

which satisfy the commutation relation,

$$[p, x] = -i\hbar\,, \tag{3.40}$$

and then solve the resulting ordinary differential equation, insisting on solutions which decrease properly at large distances. The eigenvalues can be determined in a more elegant way, however, by operator or matrix algebra techniques.

This method derives great importance from its central role in quantum field theory. Consider the operators:

$$a \equiv (xm\omega + ip)\,(2m\omega\hbar)^{-1/2} \tag{3.41}$$

$$a^\dagger \equiv (xm\omega - ip)\,(2m\omega\hbar)^{-1/2}\,. \tag{3.42}$$

The definitions of these operators differ widely in the literature. Their product is related to the Hamiltonian, so there is considerable freedom in the disposition of constant multipliers and phase factors. The use of the dagger notation is in accord with later definitions. The product of $a^\dagger$ and a is

$$\begin{aligned} a^\dagger a &= (2m\omega\hbar)^{-1}(xm\omega - ip)(xm\omega + ip) \\ &= (2m\omega\hbar)^{-1}[m^2\omega^2x^2 + p^2 - im\omega(px - xp)] \\ &= (\hbar\omega)^{-1}(H - (i\omega/2)[p, x]) = (\hbar\omega)^{-1}(H - \hbar\omega/2)\,, \end{aligned}$$

where the commutation relation (3.40) has been used in the last step. Therefore, the Hamiltonian may be expressed in terms of $a^\dagger$ and a by

$$H = \hbar\omega(a^\dagger a + \tfrac{1}{2})\,. \tag{3.43}$$

A parallel calculation shows that

$$aa^\dagger = (\hbar\omega)^{-1}(H + \hbar\omega/2)\,. \tag{3.44}$$

Consequently,

$$[a, a^\dagger] \equiv aa^\dagger - a^\dagger a = I\,. \tag{3.45}$$

We shall also need to know the commutator of H with $a^\dagger$ and a.

$$[H, a^\dagger] = \hbar\omega[(a^\dagger aa^\dagger + \tfrac{1}{2}a^\dagger) - (a^\dagger a^\dagger a + \tfrac{1}{2}a^\dagger)] = \hbar\omega a^\dagger[a, a^\dagger] = \hbar\omega a^\dagger\,. \tag{3.46}$$

Similarly,

$$[H, a] = -\hbar\omega a\,. \tag{3.47}$$

We can see on physical grounds that the simple harmonic oscillator only has energies greater than or equal to zero. This can also be shown from the operator algebra. In Section 4.4, we shall use the fact that both the operators p and x are Hermitian and therefore have real eigenvalues to prove that their squares can have only nonnegative (real) eigenvalues. Consequently, the Hamiltonian can have only positive (or zero) real eigenvalues, since $m, \omega > 0$. Assuming this result, let $E_0 \geq 0$ denote the lowest energy which the harmonic oscillator can have, the so-called *ground state*. Let u_0 be the eigenvector belonging to E_0*. Then

$$Hu_0 = E_0u_0\,.$$

Now

$$aHu_0 = E_0au_0\,.$$

* Note that we are assuming that u_0 is the *only* eigenvector belonging to E_0. For the particular operator, a, that we are using here, this assumption is correct, as we will see below. However, this fact does *not* follow just from the commutation relations.

Using the commutation relation (3.47) for H and a, we obtain

$$aHu_0 = (Ha + \hbar\omega a)u_0 = E_0 a u_0 ,$$

or

$$Hau_0 = (E_0 - \hbar\omega)au_0 .$$

This is in the form of an eigenvalue equation with a new eigenvalue, $E_0 - \hbar\omega$, and a new eigenvector, (au_0). But since by assumption E_0 is the lowest allowed eigenvalue, this can only be a trivial solution to the eigenvalue equation; that is, $au_0 \equiv 0$. If we operate on this equation with $a^\dagger$, we obtain

$$a^\dagger a u_0 = (\hbar\omega)^{-1}(H - \hbar\omega/2)u_0 = 0 ,$$

or

$$Hu_0 = (\hbar\omega/2)u_0 .$$

Therefore $E_0 = \hbar\omega/2$ is the energy of the ground state, the lowest allowed energy level. To get the other energy levels, we operate on $Hu_0 = E_0 u_0$ with $a^\dagger$.

$$a^\dagger H u_0 = (Ha^\dagger - \hbar\omega a^\dagger)u_0 = E_0 a^\dagger u_0 ,$$

or

$$H(a^\dagger u_0) = (E_0 + \hbar\omega)a^\dagger u_0 .$$

That is, $(E_0 + \hbar\omega)$ is an energy eigenvalue of H corresponding to the eigenvector $a^\dagger u_0$. If we again operate on this equation with $a^\dagger$, we get

$$a^\dagger H a^\dagger u_0 = (E_0 + \hbar\omega)(a^\dagger)^2 u_0 .$$

Using the commutation relations between $a^\dagger$ and H, we get

$$(Ha^\dagger - \hbar\omega a^\dagger)(a^\dagger u_0) = (E_0 + \hbar\omega)(a^\dagger)^2 u_0 ,$$

or

$$H(a^\dagger)^2 u_0 = (E_0 + 2\hbar\omega)(a^\dagger)^2 u_0 .$$

Thus $(E_0 + 2\hbar\omega)$ is an energy eigenvalue of H corresponding to the eigenvector $a^{\dagger 2}u_0$. By operating n times with $a^\dagger$, one obtains

$$H(a^\dagger)^n u_0 = (E_0 + n\hbar\omega)(a^\dagger)^n u_0 .$$

Successive application of $a^\dagger$ creates *all* the eigenvalues and eigenvectors (Problem 23). For this reason, $a^\dagger$ is called the *creation operator*. Calling the nth eigenvalue E_n, and the nth eigenvector u_n, we have

$$E_n = (n + \tfrac{1}{2})\hbar\omega , \qquad \text{and} \qquad u_n = c_n (a^\dagger)^n u_0 , \tag{3.48}$$

where c_n is an arbitrary constant. Note that $c_0 = 1$.

We can solve for u_0 explicitly by using $au_0 = 0$, and making the usual substitutions (3.39) for p and x. Thus

$$au_0 = (2m\omega\hbar)^{-1/2}[xm\omega + \hbar(d/dx)]u_0 = 0 ,$$

or

$$\frac{du_0}{dx} + \frac{m\omega}{\hbar} x u_0 = 0 .$$

This equation is easily solved, by one integration, to give

$$u_0 = N_0 e^{-(m\omega/2\hbar)x^2},$$

where the constant of integration, N_0, is determined by the normalization condition:

$$1 = \int_{-\infty}^{\infty} |u_0|^2\, dx = |N_0|^2 (\hbar\pi/m\omega)^{1/2}.$$

Therefore

$$N_0 = (m\omega/\hbar\pi)^{1/4},$$

and u_0 is completely determined:

$$u_0 = (m\omega/\hbar\pi)^{1/4} e^{-(m\omega/2\hbar)x^2}.$$

The other eigenfunctions can be derived from this by applying Eq. (3.48). Even the constants c_n may be found by a purely algebraic technique; we shall prove in Section 4.4 that

$$c_n = (1/n!)^{1/2}. \tag{3.49}$$

For completeness, we mention that the eigenfunctions generated by applying Eq. (3.48) are proportional to the Hermite polynomials, $H_n(y)$, times $\exp(-y^2/2)$. That is,

$$u_n(x) = N_n H_n(\alpha x) e^{-\alpha^2 x^2/2}, \tag{3.50}$$

where

$$\alpha^2 = m\omega/\hbar, \qquad \text{and} \qquad N_n = \left(\frac{\alpha}{\pi^{1/2} 2^n n!}\right)^{1/2}.$$

The first three Hermite polynomials are

$$H_0(y) = 1, \qquad H_1(y) = 2y, \qquad H_2(y) = 4y^2 - 2.$$

We have not *proved* here that the solution can be expressed in terms of Hermite polynomials. In fact, one may more easily arrive at this result by putting Eq. (3.38) in the form of a second-order differential equation and solving it using the standard method of expansion in a power series.

To complete this example, we discuss the effect of the operator, a, on the eigenfunction u_n. Remembering that $au_0 = 0$, and writing $(a^\dagger)^n$ as $a^{\dagger n}$, we have

$$\begin{aligned}
au_n &= c_n a a^{\dagger n} u_0 \\
&= c_n\{[a, a^\dagger] a^{\dagger n-1} u_0 + a^\dagger a a^{\dagger n-1} u_0\} \\
&= c_n\{a^{\dagger n-1} u_0 + a^\dagger a a^{\dagger n-1} u_0\} \\
&= c_n\{a^{\dagger n-1} u_0 + a^\dagger [a, a^\dagger] a^{\dagger n-2} u_0 + a^\dagger (a^\dagger a) a^{\dagger n-2} u_0\} \\
&= c_n\{a^{\dagger n-1} u_0 + a^{\dagger n-1} u_0 + a^{\dagger 2} a a^{\dagger n-2} u_0\} \\
&= c_n\{2a^{\dagger n-1} + a^{\dagger 2} a a^{\dagger n-2}\} u_0.
\end{aligned}$$

By continuing this process n times, we finally get

$$au_n = c_n(n a^{\dagger n-1} u_0 + a^{\dagger n} a u_0) = c_n n a^{\dagger n-1} u_0,$$

so that

$$au_n = n\frac{c_n}{c_{n-1}}(c_{n-1}a^{\dagger n-1}u_0) = n\frac{c_n}{c_{n-1}}u_{n-1} = n^{1/2}u_{n-1}\,. \tag{3.51}$$

We may also calculate the effect of $a^\dagger$ on u_n. Using Eqs. (3.43) and (3.48), we get

$$Hu_n = \hbar\omega(a^\dagger a + \tfrac{1}{2})u_n = \hbar\omega(n + \tfrac{1}{2})u_n\,.$$

Therefore

$$a^\dagger a u_n = nu_n\,,$$

so using Eq. (3.51), we obtain

$$a^\dagger(n^{1/2}u_{n-1}) = nu_n\,,$$

which, letting $n \to (n+1)$, becomes

$$a^\dagger u_n = (n+1)^{1/2}u_{n+1}\,. \tag{3.52}$$

Because of Eqs. (3.51) and (3.52), it is customary to call $a^\dagger$ and a, *raising* and *lowering* operators, or *creation* and *annihilation* operators, respectively.

3.11 THE KRONECKER PRODUCT

In this section, we want to discuss a method of combining linear operators which act on different vector spaces. We have already seen how to form the "product" of two linear operators, B_1 and B_2, which act on the same vector space U. Suppose now that we have a linear operator B acting on an n-dimensional vector space U, which takes vectors in U into vectors in U. In other words, B maps U into U, or symbolically,

$$B\colon\ U \to U\,.$$

Similarly, let A act on the m-dimensional vector space V:

$$A\colon\ V \to V\,.$$

Another important class of operators is the collection of linear operators which map U into V in a manner which preserves linear operations. That is, if C is some such operator, we want

$$C(\alpha x_1 + \beta x_2) = \alpha Cx_1 + \beta Cx_2\,.$$

Note that since $x_1, x_2 \in U$, and $Cx_1, Cx_2 \in V$, U and V must be vector spaces over the same field of scalars for this equation to make sense.

Just as we did for operators which map U into itself, we can represent operators like C by an array of scalars in the following manner. Let $\{u_i\}$ be a basis in U and $\{v_i\}$ be a basis in V. Then Cu_j is in V and must be a linear combination of the v_i. We write therefore

$$Cu_j = \sum_{k=1}^{m} c_{kj}v_k\,, \qquad j = 1, 2, \cdots, n\,.$$

Note that the array c_{kj} is in general not square, but rather has m rows and n columns. Thus if x is a vector in U such that

$$x = \sum_{j=1}^{n} \alpha_j u_j ,$$

and y is a vector in V such that

$$y = \sum_{i=1}^{m} \beta_i v_i ;$$

then, if $y = Cx$, the β's and α's are related by

$$\beta_i = \sum_{j=1}^{n} c_{ij}\alpha_j , \qquad i = 1, 2, \cdots, m .$$

We now want to consider this collection of linear operators which maps U into V as a *vector space*. The vectors in this new vector space are operators. Now C is represented by an $m \times n$ matrix, having (mn) elements. Therefore, our vector space will be mn-dimensional, since every $m \times n$ matrix can be written as a linear combination of the mn matrices

$$P_{11} = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \\ \vdots & & & \end{bmatrix}, \quad P_{12} = \begin{bmatrix} 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & \\ \vdots & & & \end{bmatrix}, \quad \text{etc.},$$

or, in other words, the mn matrices P_{ij} defined by

$$(P_{ij})_{kl} = \delta_{ik}\delta_{jl} . \tag{3.53}$$

These are linearly independent, and span the space. Thus they form a basis for the vector space of $m \times n$ matrices. Since there are mn of them, we have an mn-dimensional vector space, which we shall denote by $\mathscr{L}$. [In forming this vector space we make the logical linearity requirements, such as $(A + B)x = Ax + Bx$ for any $x \in U$, etc.]

For example, consider the following 2×3 matrix which maps from a three-dimensional vector space into a two-dimensional one:

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} .$$

In this case, $\mathscr{L}$ is six-dimensional, and if we use the basis suggested above, we may write

$$\{C\} = 1 \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 2 \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 3 \times \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$
$$+ 4 \times \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} + 5 \times \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} + 6 \times \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where we adopt the convention of bracketing an element of $\mathscr{L}$ when considered as a "vector" rather than as an "operator." In this case, $\{C\}$ can be represented

in this particular (but very natural) basis by

$$\{C\} \to \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}.$$

With the preliminary definitions and discussion out of the way, it is a simple matter to define the *Kronecker* (or *direct*) *product*. Let A and B be operators defined on V and U, which respectively map V and U into themselves, and let C be some operator which maps U into V. For our given A and B, there is another operator that maps U into V which one would want to associate naturally with C, namely

$$C' = AC\tilde{B}\,.$$

That is, $\tilde{B}$ maps U into U, C maps U into V, and finally, A maps V into V. The reason we use $\tilde{B}$ rather than B is to make the properties of the Kronecker product as symmetrical as possible.

Now consider $\{C\}$ and $\{C'\}$ as elements of $\mathscr{L}$. We define the Kronecker product of A and B, denoted by $A \otimes B$, to be that operator on $\mathscr{L}$ which takes $\{C\}$ into $\{C'\}$, that is,

$$(A \otimes B)\{C\} = \{C'\} = \{AC\tilde{B}\}\,, \tag{3.54}$$

for all $C \in \mathscr{L}$.

The properties of the Kronecker product are easy to derive from Eq. (3.54). We have immediately

$$I \otimes I = I\,, \tag{3.55}$$

where by I we mean the unit operator in whatever dimensional space is appropriate. Other immediate consequences of Eq. (3.54) are

$$(A_1 + A_2) \otimes B = A_1 \otimes B + A_2 \otimes B \tag{3.56}$$

and

$$A \otimes (B_1 + B_2) = A \otimes B_1 + A \otimes B_2\,. \tag{3.57}$$

Now we consider

$$\begin{aligned}(A_1 \otimes B_1)(A_2 \otimes B_2)\{C\} &= (A_1 \otimes B_1)\{A_2C\tilde{B}_2\} \\ &= \{A_1A_2C\tilde{B}_2\tilde{B}_1\} \\ &= (A_1A_2C\widetilde{B_1B_2}) \\ &= (A_1A_2 \otimes B_1B_2)\{C\}\,,\end{aligned}$$

so we have established the important property

$$(A_1 \otimes B_1)(A_2 \otimes B_2) = (A_1A_2 \otimes B_1B_2)\,, \tag{3.58}$$

where $A_1, A_2 \in V$, and $B_1, B_2 \in U$. The fact that the right-hand side of (3.58) is $A_1A_2 \otimes B_1B_2$ rather than $A_1A_2 \otimes B_2B_1$ is the reason for our use of $\tilde{B}$ rather than B in our definition of the Kronecker product. We leave it to the reader to show that if both A^{-1} and B^{-1} exist, then

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1} . \tag{3.59}$$

Finally, let us see what $A \otimes B$ looks like when represented as a matrix in the basis of Eq. (3.53). There is a minor notational annoyance which we should first point out: Our basis "vectors," the P_{ij}, are indexed by two subscripts instead of the usual single subscript that we are used to seeing. This is because we are perversely regarding operators (matrices) as vectors. Thus we now have to talk about the ij-component of a *vector*! Likewise, we shall have to refer to the (ij), (kl)-component of a matrix of an operator like $A \otimes B$.

The (ij), (kl) element of $A \otimes B$ is found in the usual way. We write

$$(A \otimes B)\{P_{kl}\} = \sum_{i,j} [A \otimes B]_{ij,\,kl}\{P_{ij}\} ,$$

where we denote the (ij), (kl) element of $A \otimes B$ by the symbol $[A \otimes B]_{ij,\,kl}$. But

$$(A \otimes B)\{P_{kl}\} = \{AP_{kl}\tilde{B}\} ,$$

by Eq. (3.54), so we obtain the equation

$$\{AP_{kl}\tilde{B}\} = \sum_{i,j} [A \otimes B]_{ij,\,kl}\{P_{ij}\} .$$

In other words, the number $[A \otimes B]_{ij,\,kl}$ is just the ijth element of the *matrix* $AP_{kl}\tilde{B}$, that is,

$$[A \otimes B]_{ij,\,kl} = \sum_{\mu,\,\nu} A_{i\mu}(P_{kl})_{\mu\nu}\tilde{B}_{\nu j} .$$

But by Eq. (3.53), we get

$$\begin{aligned}[A \otimes B]_{ij,\,kl} &= \sum_{\mu,\,\nu} A_{i\mu}\delta_{k\mu}\delta_{l\nu}\tilde{B}_{\nu j} \\ &= A_{ik}\tilde{B}_{lj} .\end{aligned}$$

Since $\tilde{B}_{lj} = B_{jl}$, we get finally

$$[A \otimes B]_{ij,\,kl} = A_{ik}B_{jl} , \tag{3.60}$$

so the elements of $A \otimes B$ are obtained easily from the elements of A and the elements of B.

Note that the trace of $A \otimes B$ is just

$$\operatorname{tr}(A \otimes B) = \sum_{i,j} [A \otimes B]_{ij,\,ij} = \sum_{i,j} A_{ii}B_{jj} = (\operatorname{tr} A)(\operatorname{tr} B) ,$$

or

$$\operatorname{tr}(A \otimes B) = (\operatorname{tr} A)(\operatorname{tr} B) . \tag{3.61}$$

This is a relation which we shall find very useful when we discuss group theory.

It is instructive to see how these things look written out in matrix form. To do this we have to adopt an ordering for the subscripts (ij). We shall write

the column vectors, which represent elements of $\mathscr{L}$, in the following order:

$$\begin{bmatrix} C_{11} \\ C_{12} \\ \vdots \\ C_{1n} \\ C_{21} \\ C_{22} \\ \vdots \\ C_{2n} \\ \vdots \\ C_{m1} \\ C_{m2} \\ \vdots \\ C_{mn} \end{bmatrix}$$

Hence our matrices, which represent objects like $A \otimes B$, will take the form

$$\begin{bmatrix} [A \otimes B]_{11,11} & [A \otimes B]_{11,12} & \cdots & [A \otimes B]_{11,1n} & [A \otimes B]_{11,21} & \cdots & [A \otimes B]_{11,mn} \\ [A \otimes B]_{12,11} & [A \otimes B]_{12,12} & \cdots & [A \otimes B]_{12,1n} & [A \otimes B]_{12,21} & \cdots & [A \otimes B]_{12,mn} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ [A \otimes B]_{1n,11} & [A \otimes B]_{1n,12} & \cdots & [A \otimes B]_{1n,1n} & [A \otimes B]_{1n,21} & \cdots & [A \otimes B]_{1n,mn} \\ [A \otimes B]_{21,11} & [A \otimes B]_{21,12} & \cdots & [A \otimes B]_{21,1n} & [A \otimes B]_{21,21} & \cdots & [A \otimes B]_{21,mn} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ [A \otimes B]_{mn,11} & [A \otimes B]_{mn,12} & \cdots & [A \otimes B]_{mn,1n} & [A \otimes B]_{mn,21} & \cdots & [A \otimes B]_{mn,mn} \end{bmatrix}.$$

Using Eq. (3.60), we obtain

$$\begin{bmatrix} A_{11}B_{11} & A_{11}B_{12} & \cdots & A_{11}B_{1n} & A_{12}B_{11} & \cdots & A_{12}B_{1n} & \cdots & A_{1m}B_{1n} \\ A_{11}B_{21} & A_{11}B_{22} & \cdots & A_{11}B_{2n} & A_{12}B_{21} & \cdots & A_{12}B_{2n} & \cdots & A_{1m}B_{2n} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & & \vdots \\ A_{11}B_{n1} & A_{11}B_{n2} & \cdots & A_{11}B_{nn} & A_{12}B_{n1} & \cdots & A_{12}B_{nn} & \cdots & A_{1m}B_{nn} \\ A_{21}B_{11} & A_{21}B_{12} & \cdots & A_{21}B_{1n} & A_{22}B_{11} & \cdots & A_{22}B_{1n} & \cdots & A_{2m}B_{1n} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & & \vdots \\ A_{21}B_{n1} & A_{21}B_{n2} & \cdots & A_{21}B_{nn} & A_{22}B_{n1} & \cdots & A_{22}B_{nn} & \cdots & A_{2m}B_{nn} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & & \vdots \\ A_{m1}B_{n1} & A_{m1}B_{n2} & \cdots & A_{m1}B_{nn} & A_{m2}B_{n1} & \cdots & A_{m2}B_{nn} & \cdots & A_{mm}B_{nn} \end{bmatrix}.$$

Inspecting this result, we see that if we break the matrix representative of $A \otimes B$

into $n \times n$ blocks (there will be m^2 of them) starting in the upper left corner, then the matrix of $A \otimes B$ can be written symbolically as

$$A \otimes B \to \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ \vdots & & & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mm}B \end{bmatrix}.$$

That is, to form the matrix of $A \otimes B$, we replace the ij-element of the matrix of A, A_{ij}, by $A_{ij}B$, that is, by an $n \times n$ array. Note that in this form the theorem on traces [Eq. (3.61)] is particularly easy to see.

As an example of this kind of matrix, we consider the 2×3 matrix

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

which, when considered as a vector in $\mathscr{L}$ may be represented by

$$\{C\} \to \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}.$$

Now suppose that

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}, \qquad \text{so} \qquad \tilde{B} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix},$$

and

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}.$$

B maps U into itself and A maps V into itself. Thus

$$C' = AC\tilde{B} = \begin{bmatrix} 4 & 7 & 10 \\ 24 & 45 & 60 \end{bmatrix}$$

so that

$$\{C'\} \to \begin{bmatrix} 4 \\ 7 \\ 10 \\ 24 \\ 45 \\ 60 \end{bmatrix}.$$

The matrix of $A \otimes B$ is easily computed, either from Eq. (3.60) or by the picture given above. We get

$$A \otimes B \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 3 & 1 & 2 & 6 & 2 \end{bmatrix}$$

and we leave it to the reader to check that, indeed,

$$(A \otimes B)\{C\} = \{C'\}\,.$$

PROBLEMS

1. Prove that the n complex nth roots of unity together with multiplication of complex numbers form an abelian group.
   [*Hint*: From De Moivre's formula, the n complex nth roots of unity are given by
   $$e_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n} = e^{2ik\pi/n}, \qquad k = 0, 1, \cdots, n-1,$$
   where $i^2 = -1$.]
2. Show that 3 vectors in three-dimensional space are linearly dependent if and only if $\mathbf{A}\cdot(\mathbf{B} \times \mathbf{C}) = 0$.
3. In the vector space of real n-tuples, prove that the n vectors
   $$\begin{aligned} x_1 &= (1, 1, \cdots, 1, 1), \\ x_2 &= (0, 1, \cdots, 1, 1), \\ x_3 &= (0, 0, 1, \cdots, 1, 1), \\ &\cdot \\ &\cdot \\ &\cdot \\ x_n &= (0, 0, \cdots, 0, 1) \end{aligned}$$
   are linearly independent.
4. Prove that any set of $n + 1$ vectors in an n-dimensional vector space is linearly dependent.
5. Prove that if the set of n-vectors $\{x_i\}$ is a basis for a vector space V, then the set $\{c_i x_i\}$ is also a basis for V, where the c_i are arbitrary nonzero scalars. Interpret this geometrically.
6. Prove that a basis is a maximal linearly independent set of vectors.
7. a) Prove that any two $n \times n$ diagonal matrices commute and that their product is also a diagonal matrix.
   b) If A commutes with all diagonal matrices B, then A is diagonal.
8. Prove that if the matrix A is idempotent and A is not the unit matrix, then A is singular.
9. Define the commutator $[A, B]$ of two matrices by $[A, B] = AB - BA$, and the anticommutator $\{A, B\}$ by $\{A, B\} = AB + BA$. Show that

a) $[A, B] = -[B, A]$, b) $[AB, C] = A[B, C] + [A, C]B$, c) $[AB, C] = A\{B, C\} - \{A, C\}B$, d) $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$.

10. Show that if a linear transformation A is known to have a *unique* left inverse (that is $BA = I$), then B is also a right inverse (that is $AB = I$).
[*Hint*: Consider the linear transformation $C = AB - I + B$, which you suspect (but must prove) equals B if the theorem is true, etc.]
11. Prove that any antisymmetric matrix of *odd order* is singular. The "odd order" qualification is essential; the Pauli spin matrix

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

is a second-order antisymmetric matrix, but since $\sigma_y^2 = I$, σ_y *does have* an inverse, namely $\sigma_y^{-1} = \sigma_y$.
12. The Pauli spin matrices are defined as follows:

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

where $i = \sqrt{-1}$. Prove that

$$\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij} \qquad \text{and} \qquad \sigma_i\sigma_j - \sigma_j\sigma_i = 2i\sum_{k=1}^{3} \epsilon_{ijk}\sigma_k .$$

Thus the Pauli matrices anticommute.
13. Prove that tr (AB) = tr (BA) for arbitrary $n \times n$ matrices, A and B. Use this result to prove that if C and D are similar, then tr (C) = tr (D).
14. Let A be an $n \times n$ matrix with real entries. Prove that $A = 0$ if and only if tr $(\tilde{A}A) = 0$.
15. Let A be an $n \times n$ idempotent matrix (that is, $A^2 = A$).
    a) Prove that each eigenvalue of A is either 0 or 1.
    b) It can be shown that an idempotent matrix can always be diagonalized by a similarity transformation. Using this, prove that tr A is integral and $0 \leq \text{tr}\, A \leq n$.
16. Show that the eigenvalues of

$$M = \begin{bmatrix} 3 & 5 & 8 \\ -6 & -10 & -16 \\ 4 & 7 & 11 \end{bmatrix}$$

are $\lambda = 0, 1, 3$, and that the corresponding eigenvectors are

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \qquad \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \qquad \text{and} \qquad \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}.$$

Construct a diagonalizing matrix P, prove that its inverse exists, compute its inverse, and verify that $P^{-1}MP$ is diagonal with the eigenvalues on the diagonal. Note that det $M = 0$. Is it true that any diagonalizable matrix with an eigenvalue equal to zero is singular?

17. Denote the two-dimensional rotation matrix for a rotation of the coordinate axes through an angle θ by

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}.$$

Find the eigenvalues and eigenvectors of A. Find a diagonalizing matrix for A, that is, a matrix P such that $P^{-1}AP$ is a diagonal matrix. Demonstrate that P is such a matrix by inverting P and forming the product $P^{-1}AP$.

18. Show that if

$$A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$$

there exists no nonsingular matrix P of order 2 such that $P^{-1}AP = D$, where D is a diagonal matrix. This example shows that not every matrix can be diagonalized by a similarity transformation.

19. Show that the determinant of *any* matrix is equal to the product of its eigenvalues and that the trace of a matrix is equal to the sum of its eigenvalues. Thus show that if a matrix has a zero eigenvalue it is not invertible.

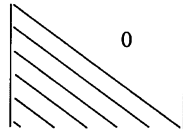

Fig. 3.4

20. Show that by adding and subtracting rows (multiples of rows) systematically, any determinant can be reduced to evaluating a determinant of the type shown in Fig. 3.4, where there are terms on the diagonal and below the diagonal, and only zeros above the diagonal. Clearly, this same process can now be applied to the columns to leave us with the trivial problem of evaluating the determinant of a diagonal matrix. However, the evaluation is already trivial after the first step. Why?
21. Derive an expression for the inverse of a matrix A in terms of the eigenvalues and eigenvectors of A. Assume that the inverse of A exists, and that the eigenvectors of A span the space in which A acts.
22. Consider an $n \times n$ matrix Q whose elements are given by $Q_{ij} = v_i v_j$, where v is a vector in n-dimensional space. Find the eigenvalues and eigenvectors of Q. Are the eigenvectors all unique?
23. Show that all eigenvalues of the quantum oscillator have the form $E_n = \hbar\omega(n + \frac{1}{2})$. [*Hint*: Assume that there is an eigenvector y and associated eigenvalue λ such that $Hy = \lambda y$ and using $[a, H] = \hbar\omega a$, create eigenvalues $\lambda - \hbar\omega$, $\lambda - 2\hbar\omega$, etc., and arrive at a contradiction.]
24. Show that if A^{-1} and B^{-1} exist, then $A^{-1} \otimes B^{-1} = (A \otimes B)^{-1}$.
25. If A and B are similar, prove that A^n and B^n are similar.
26. Show that $(C + D)^{-1} = C^{-1} - C^{-1}D(C + D)^{-1}$. Thus for D "small" we can iterate to get $(C + D)^{-1} = C^{-1} - C^{-1}DC^{-1} + C^{-1}DC^{-1}DC^{-1} \cdots$ which is the operator analogue of the familiar expansion of $(x + y)^{-1}$ when y is "small."

27. *The Dirac matrices.* Assume that for some n there exist *four* $n \times n$ matrices which satisfy the anticommutation relations $\gamma_i\gamma_j + \gamma_j\gamma_i = 2I\delta_{ij}$, where $i, j = 1, 2, 3$, or 4.
a) Prove that $\gamma_i^2 = I$ (the unit matrix), for $i = 1, 2, 3$, or 4.
b) Prove that $\det \gamma_i = \pm 1$, for $i = 1, 2, 3$, or 4.
c) If λ is an eigenvalue of any one of the γ_i, show that $\lambda = \pm 1$.
d) Show that $\operatorname{tr} \gamma_i = 0$, for $i = 1, 2, 3$, or 4.
The final part of this problem consists in showing that there exists no set of four $n \times n$ matrices of order less than 4 which satisfies the above anticommutation relations. You may, of course, do this in any way you like. One way is to prove the statement via the following two steps.
e) Show that there exists no set of four $n \times n$ matrices of odd order which satisfies the anticommutation relations.
f) Show that any *four* 2×2 matrices cannot satisfy the anticommutation relations.
[*Hint*: The three 2×2 Pauli matrices do satisfy the relations

$$\sigma_i\sigma_j + \sigma_j\sigma_i = 2I\delta_{ij}, \qquad i, j = 1, 2, 3,$$

and any 2×2 matrices can be expanded as a linear combination of the three Pauli matrices and the 2×2 unit matrix.]

(*Note*: That there exist four nonunique 4×4 matrices satisfying these relations is shown in any standard treatment of the Dirac equation. This problem shows that one need not bother looking for four matrices of lower than fourth order which obey the anticommutation relations, a point of great theoretical significance.)

28. Show that the operator

$$T = I + \frac{xD}{1!} + \frac{(xD)^2}{2!} + \cdots + \frac{(xD)^n}{n!},$$

where $D = d/dt$, acts as a translation operator on the space of polynomials (in the variable t) of degree $\leqq n$, that is,

$$Tf(t) = f(t + x),$$

if $f(t)$ is in the space of polynomials of degree n. This is the end of the problem; an application of this result follows, which is typical of various symbolic operator solutions of problems in physics (e. g. Blankenbecler's method for obtaining integrals over the Fermi-Dirac distribution function*). The present application is due largely to L. N. Hand. The result was obtained by other methods by Jacob Bernoulli (1654–1705).

We first note that in the space of continuous functions which are infinitely differentiable, the translation operator would be given by

$$T = e^{xD}.$$

Some interesting results can be obtained using this formula. For example, a general expression for $\sum_{k=0}^{k=n} k^m$ can be derived for any integers $n \geqslant 0$ and $m > 0$. Let $F_m(n) = \sum_{k=0}^{n} k^m$. Now $F_m(n) - F_m(n-1) = n^m$. Since $F_m(n)$ is a polynomial,†

* See Kittel, *Elementary Statistic Physics*, New York, Wiley 1958, § 20.
† See Stewart, *Theory of Numbers*, New York, Macmillan 1952, p. 19.

we can write $e^{-D}F_m(n) = F_m(n-1)$, and, consequently,

$$F_m(n) - F_m(n-1) = (1 - e^{-D})F_m(n) = n^m$$

or, symbolically,

$$F_m(n) = \left(\frac{1}{1-e^{-D}}\right)n^m .$$

Now,

$$\frac{1}{1-e^{-x}} = x^{-1} + \frac{1}{2} + \sum_{j=0}^{\infty} \frac{(-1)^j B_{j+1}}{[2(j+1)]!} x^{2j+1} ,$$

where the B_i's are the Bernoulli numbers

$$B_1 = \tfrac{1}{6}, \qquad B_2 = \tfrac{1}{30}, \qquad B_3 = \tfrac{1}{42}, \qquad B_4 = \tfrac{1}{30}, \quad \text{etc.}$$

(See any mathematical tables.) If we blithely use this series for $x = D$, interpreting D^{-1} as integration, and note that $D^p n^m = [m!/(m-p)!]n^{m-p}$, then we can calculate $F_m(n)$.

$$F_m(n) = \frac{n^{m+1}}{m+1} + C_m + \frac{n^m}{2} + \sum_{j=0}^{J} \frac{(-1)^j B_{j+1} m!}{[2(j+1)]!\,[m-(2j+1)]!} n^{[m-(2j+1)]},$$

where J equals $(m-1)/2$ or $(m-2)/2$, whichever is integral. The constant of integration, C_m, can be determined by using $F_m(0) = 0$. We see immediately that $C_m = 0$ if m is even and $C_m = (-1)^{(m+1)/2} B_{(m+1)/2}/(m+1)$ if m is odd; in other words C_m is chosen so that no constant term appears in $F_m(n)$. For example,

$$F_1(n) = \sum_{k=0}^{n} k = \frac{n^2}{2} + C_1 + \frac{n}{2} + \frac{B_1}{2} .$$

In order to satisfy $F_1(0) = 0$ we must choose the constant equal to $-B_1/2$, giving the usual result:

$$F_1(n) = \tfrac{1}{2}(n^2 + n) .$$

Similarly, we find the formulas:

$$F_2(n) \equiv \sum_{k=0}^{n} k^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6} = \frac{1}{6}n(n+1)(2n+1) ,$$

$$F_3(n) \equiv \sum_{k=0}^{n} k^3 = \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} = \left[\frac{n(n+1)}{2}\right]^2,$$

$$F_4(n) \equiv \sum_{k=0}^{n} k^4 = \frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{n}{30} ,$$

$$F_5(n) \equiv \sum_{k=0}^{n} k^5 = \frac{n^6}{6} + \frac{n^5}{2} + \frac{5}{12}n^4 - \frac{n^2}{12} .$$

Of course, the level of rigor in this derivation is simply appalling. However, the fact that we get away with it makes it plausible that this derivation could be made rigorous. Very often new mathematics results from the formalization of just such sloppy forays into the unknown. The theory of distributions which formalizes the symbolic functions, such as delta functions, used by physicists for fifty

years prior to their formalization, is a case in point. The present example, however, is a treatment of results long known and rigorously established. It is presented partly as an antidote to the more rigorous material covered in this and subsequent chapters.

FURTHER READINGS

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